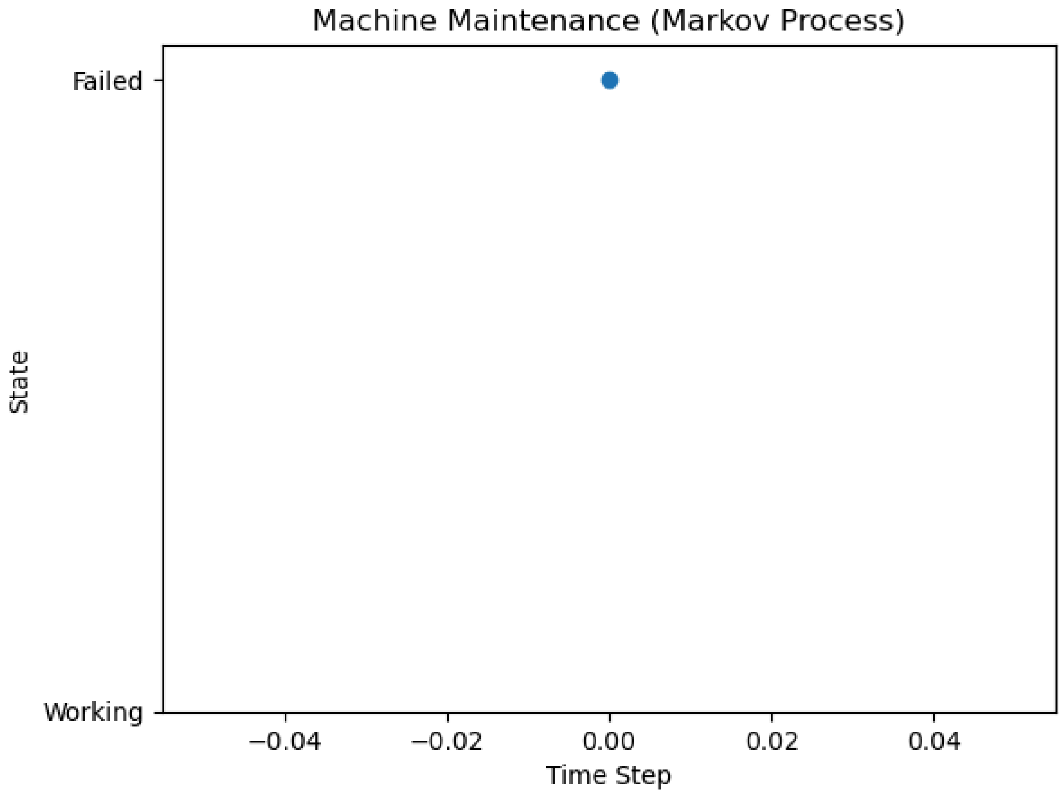
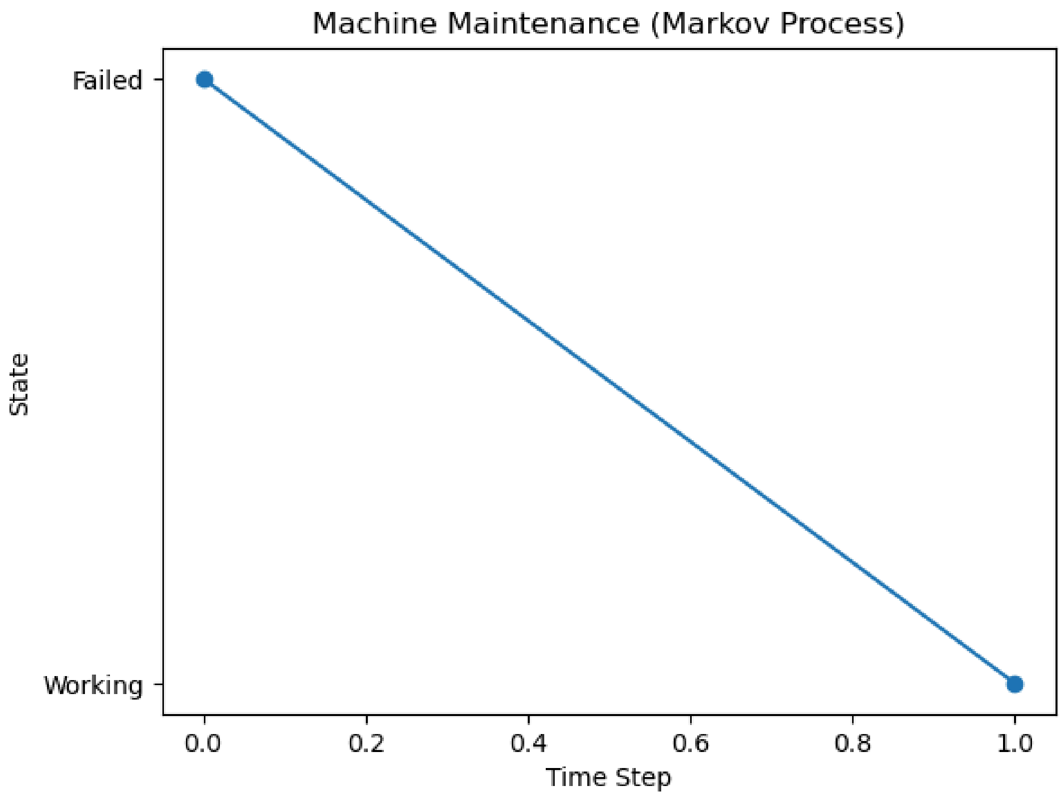
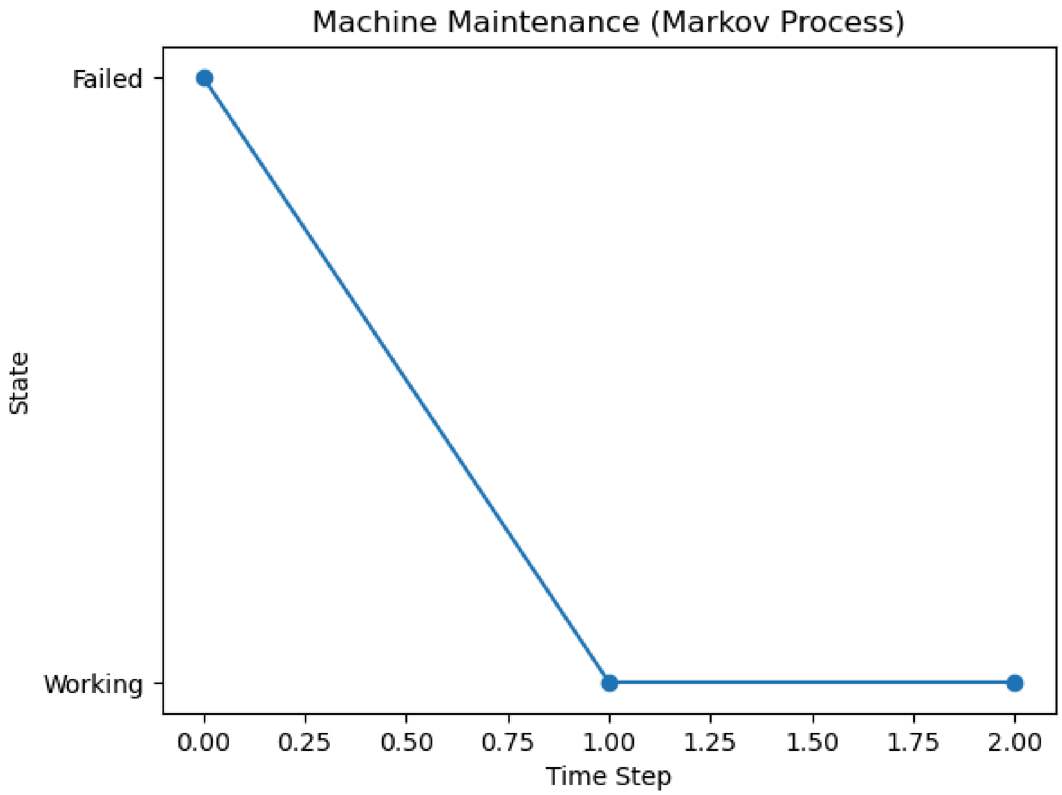
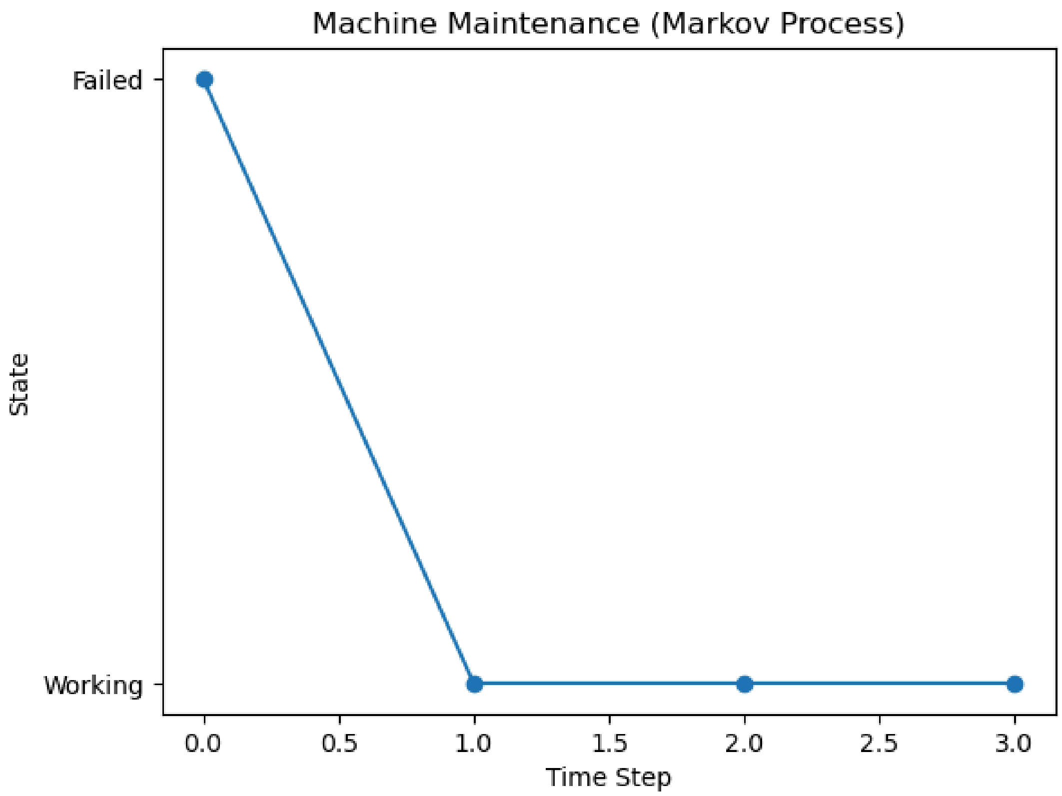
In [1]:

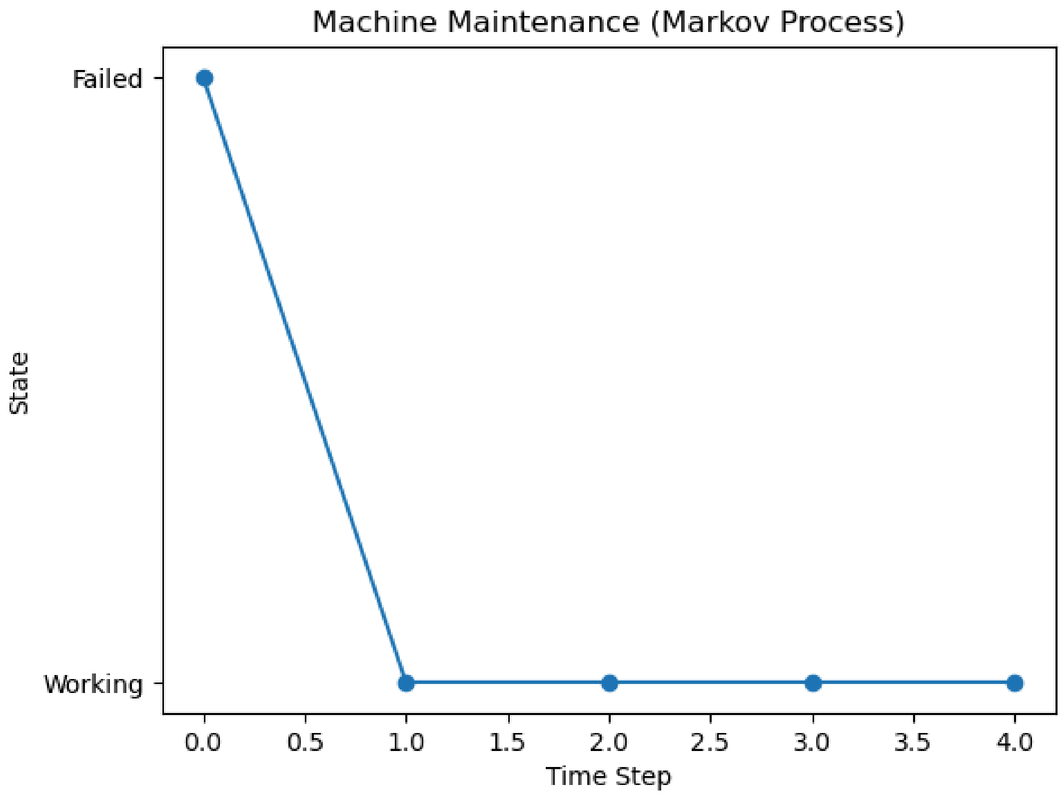
```python
import numpy as np
import matplotlib.pyplot as plt
transition_matrix = [
 [0.8, 0.2],
 [0.6, 0.4]
]
states = ["Working", "Failed"]
current_state = 0
history = []
plt.ion()
for i in range(50):
 current_state = np.random.choice([0, 1], p=transition_matrix[current_state])
 history.append(current_state)

 plt.clf()
 plt.plot(history, marker='o')
 plt.yticks([0, 1], states)
 plt.title("Machine Maintenance (Markov Process)")
 plt.xlabel("Time Step")
 plt.ylabel("State")

 plt.pause(0.2)
plt.ioff()
plt.show()
```

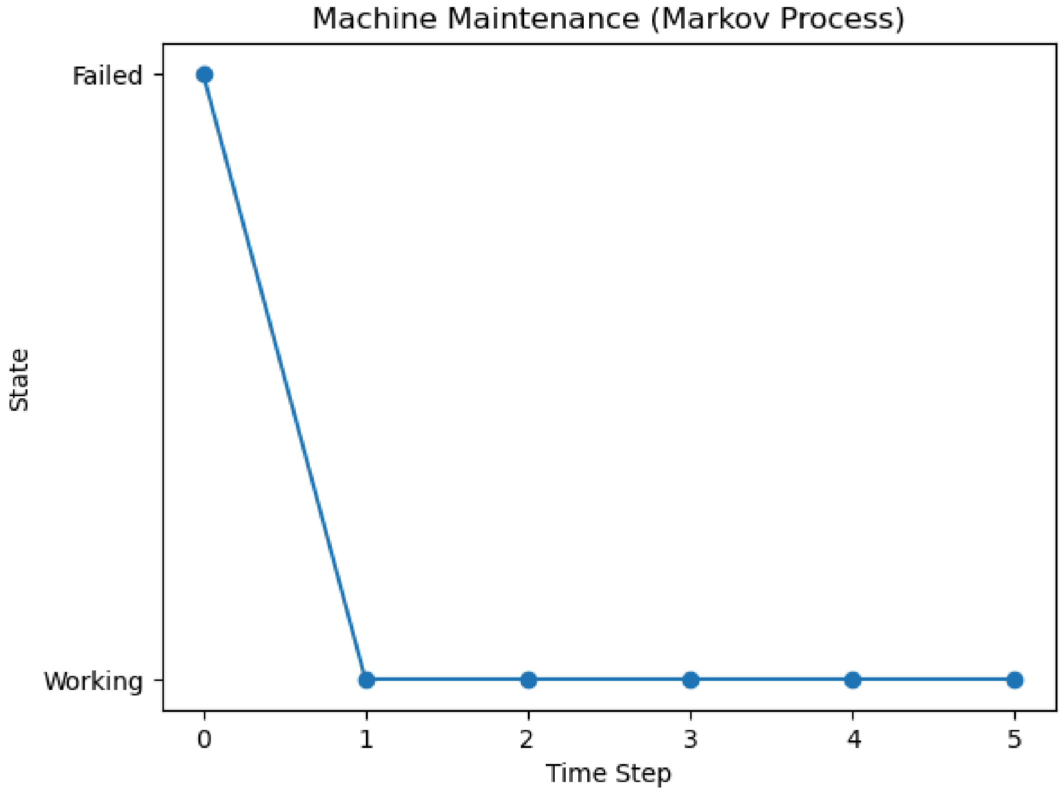


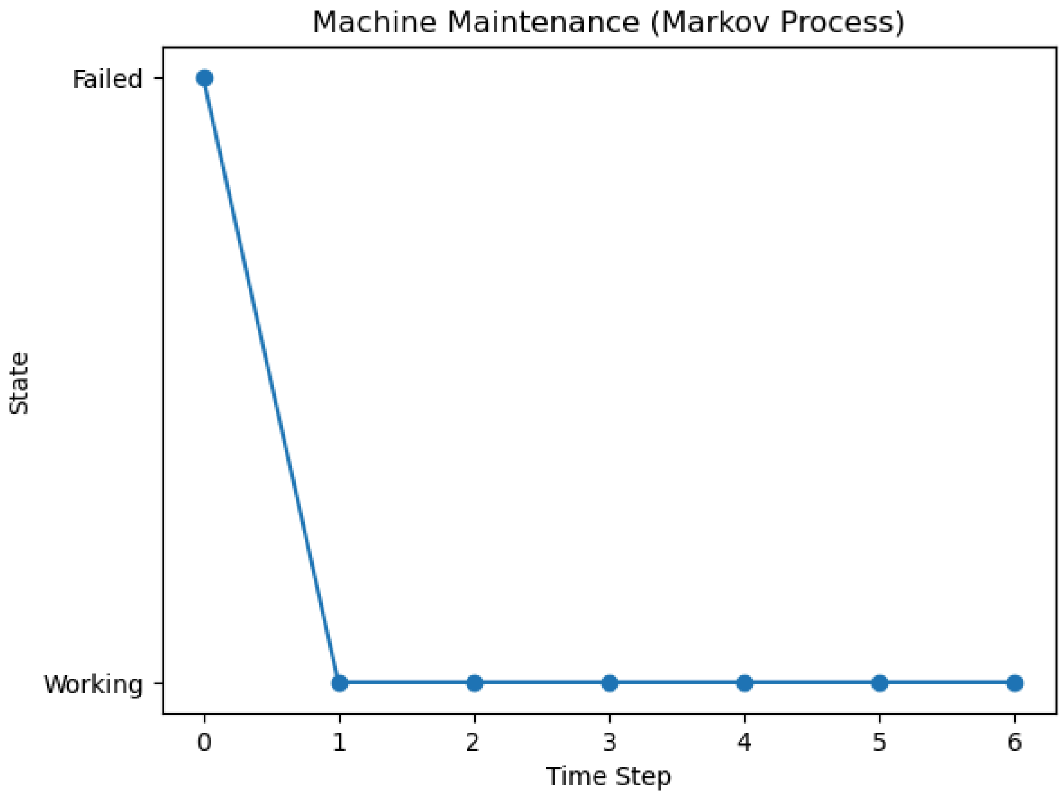

Machine Maintenance (Markov Process)
Failed
Working
State
0.0
0.2
0.4
0.6
0.8
1.0
Time Step

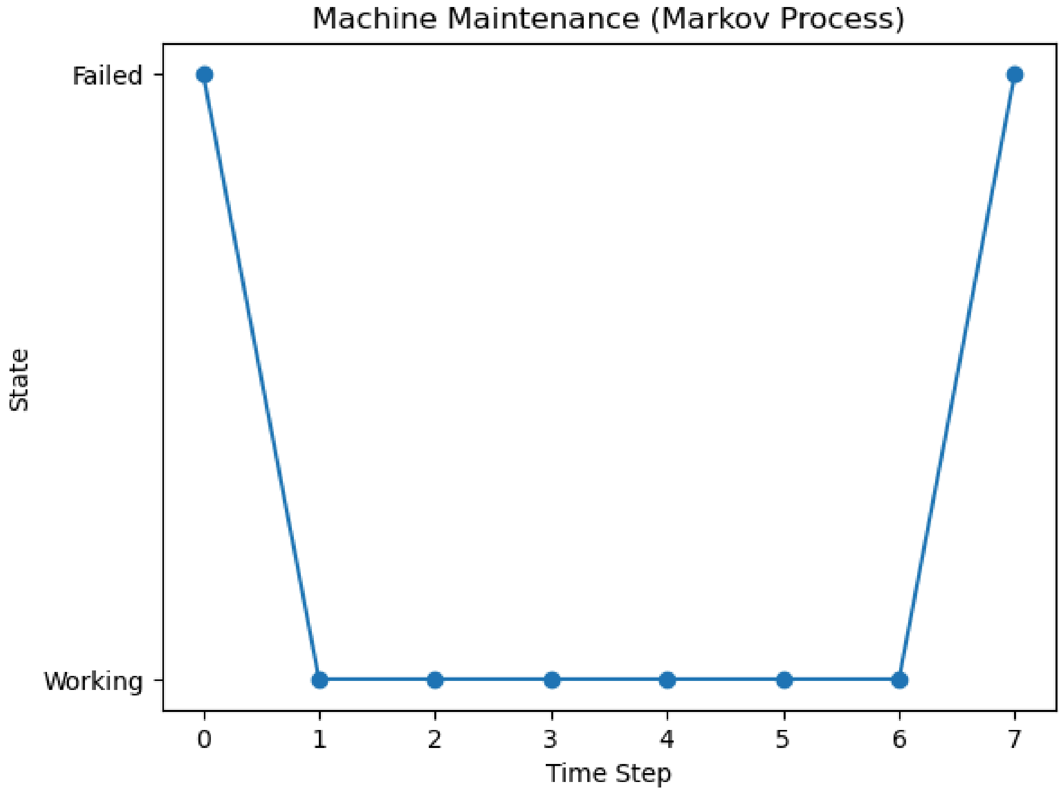

Machine Maintenance (Markov Process)
Failed
Working
State
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time Step

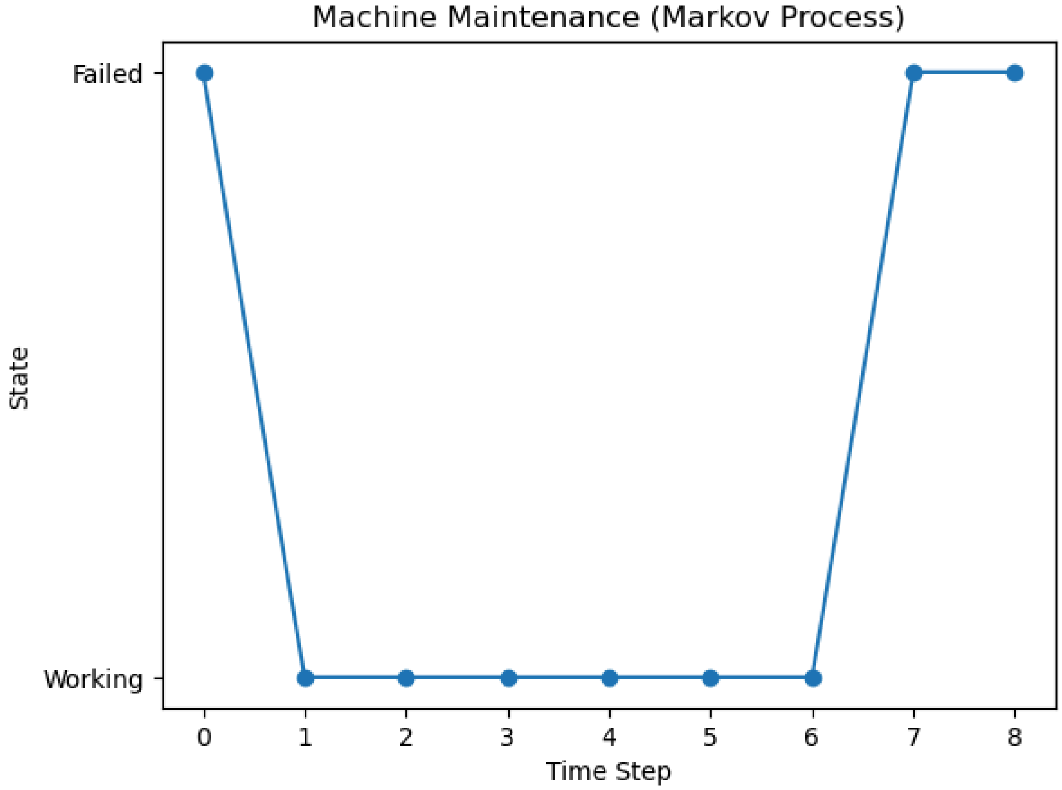

Machine Maintenance (Markov Process)
Failed
Working
State
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Time Step

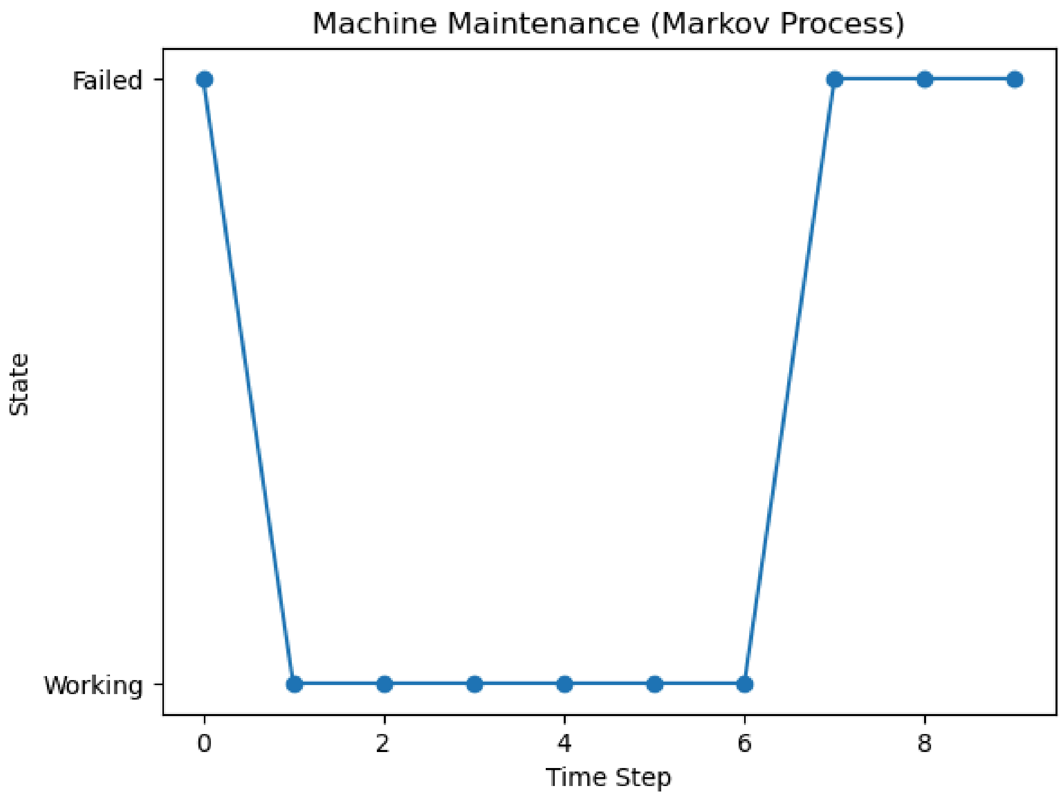

Machine Maintenance (Markov Process)
Failed
Working
State
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
1
2
3
4
5
Time Step

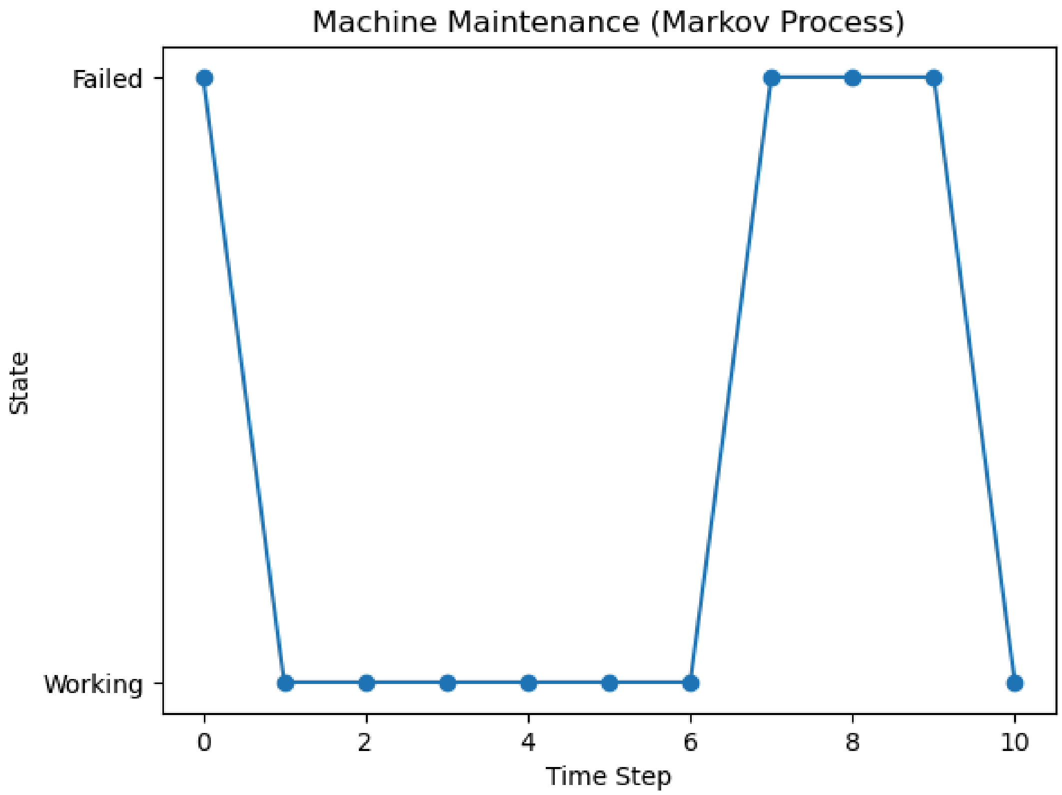

Machine Maintenance (Markov Process)
Failed
Working
State
0
1
2
3
4
5
6
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
1
2
3
4
5
6
7
Time Step

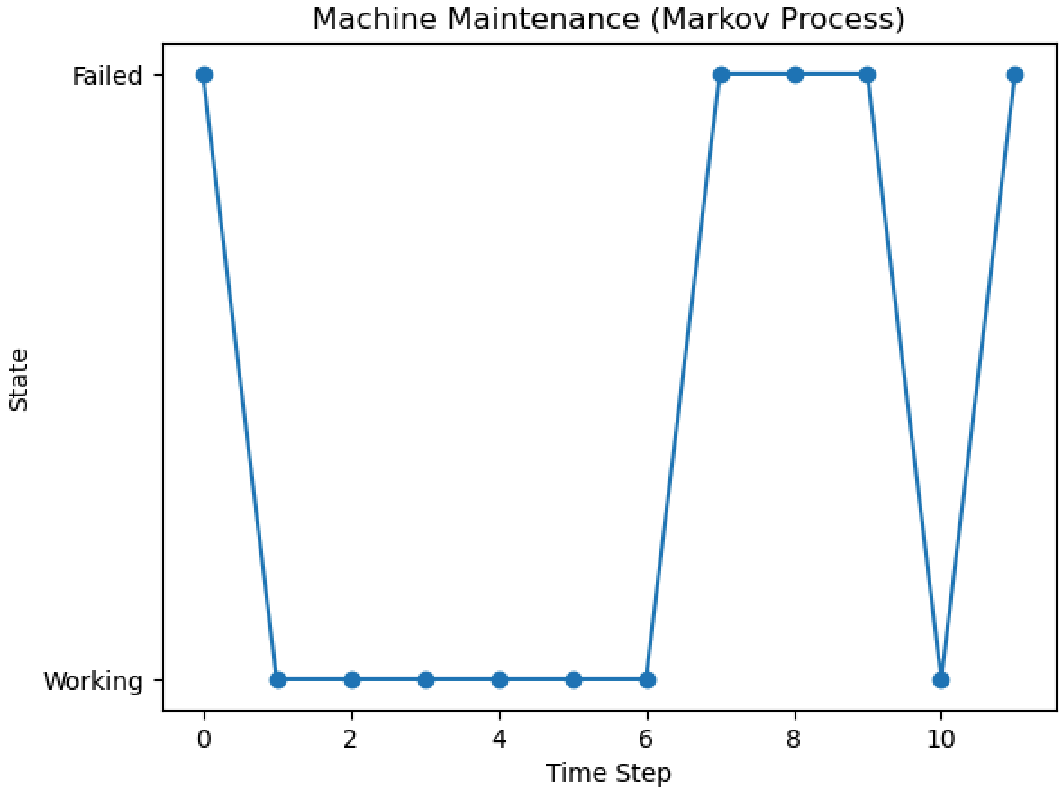

Machine Maintenance (Markov Process)
Failed
Working
State
0
1
2
3
4
5
6
7
8
Time Step

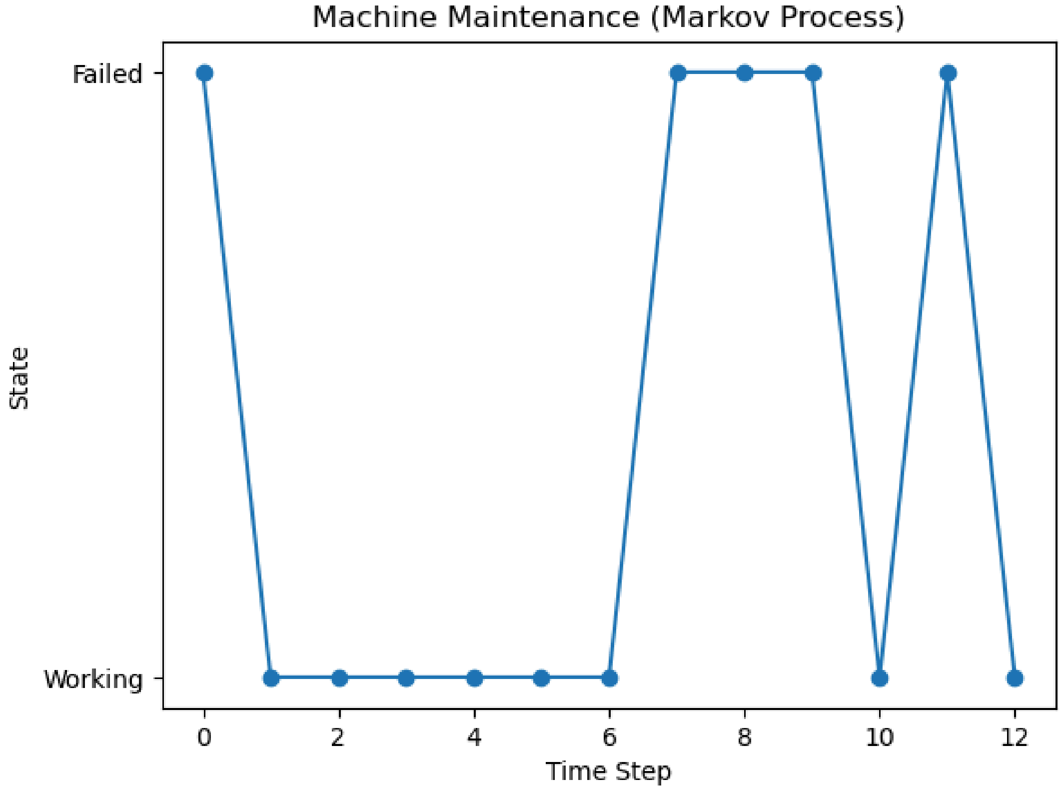


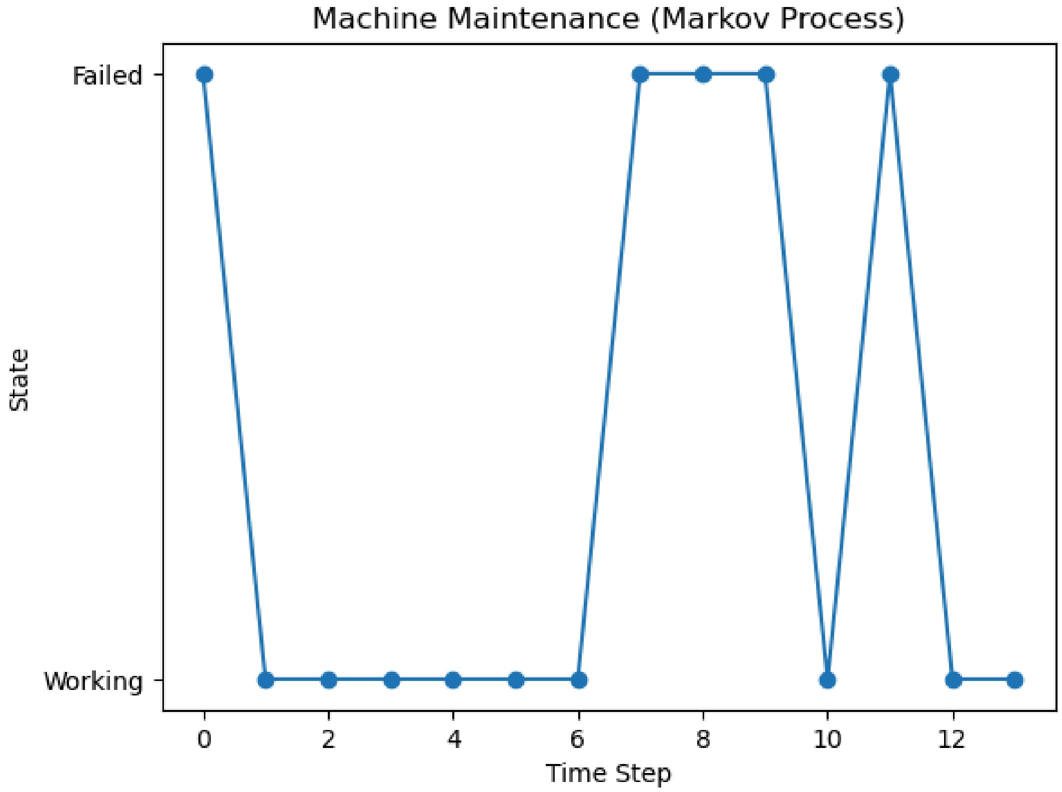
Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
Time Step

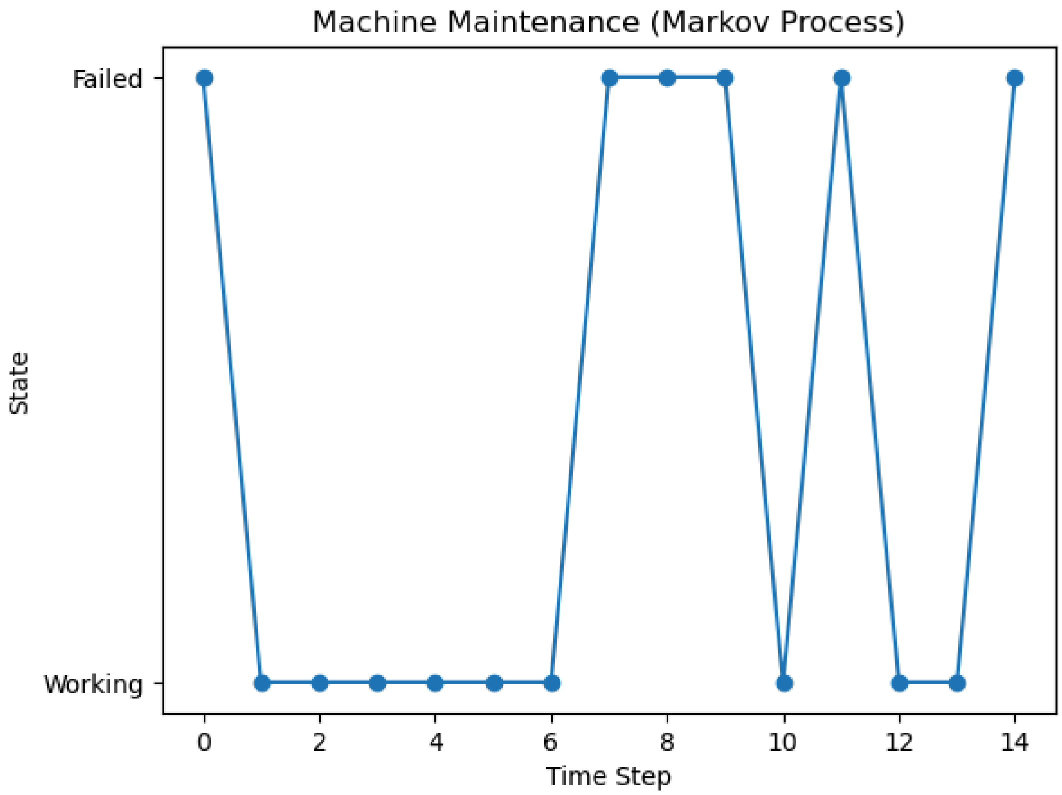


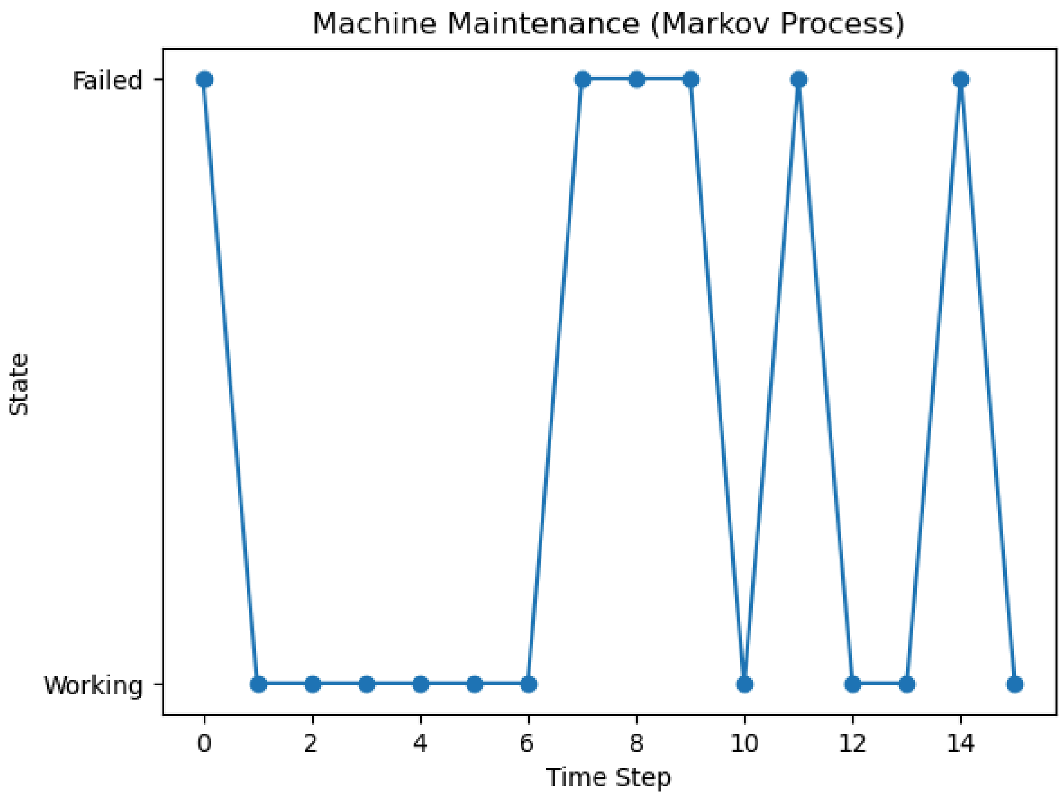
Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
12
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
12
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
12
14
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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4
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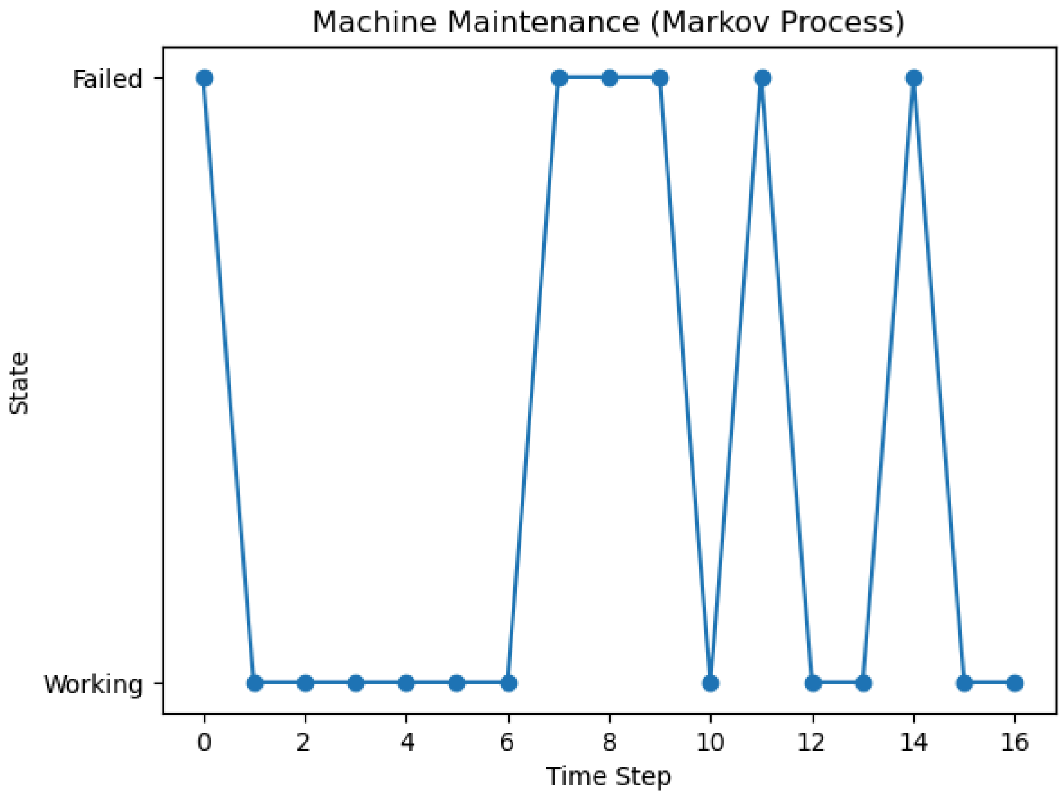
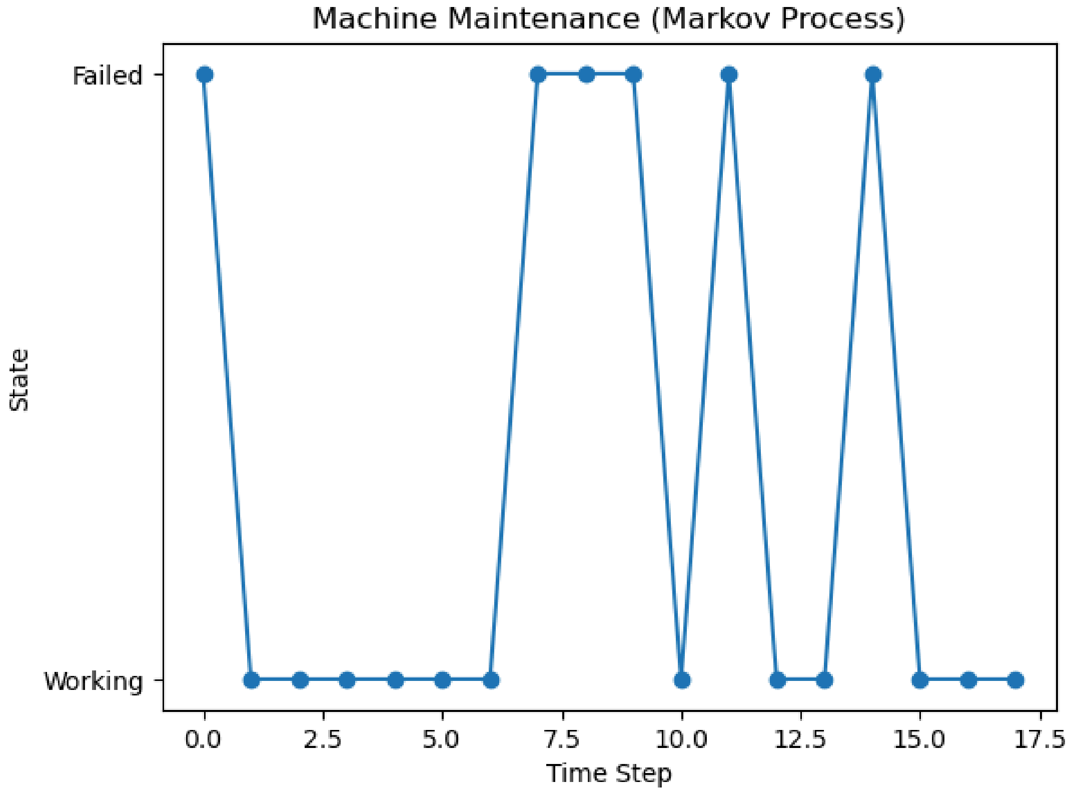
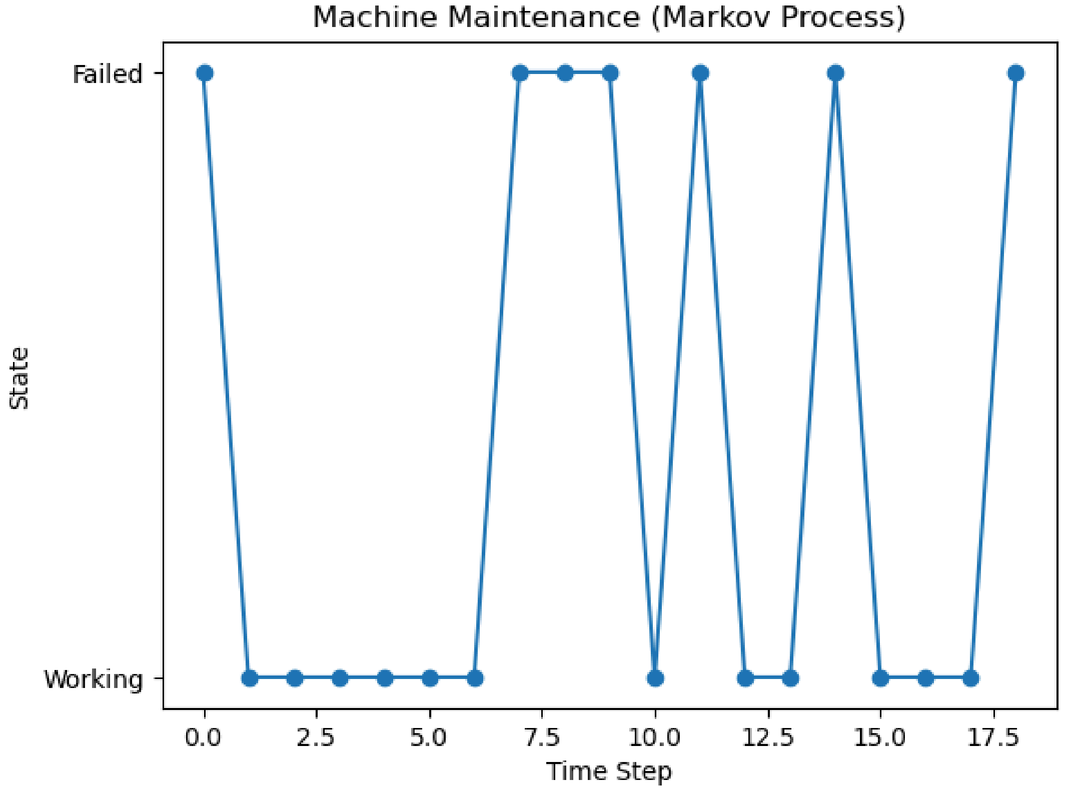
8
10
12
14
Time Step

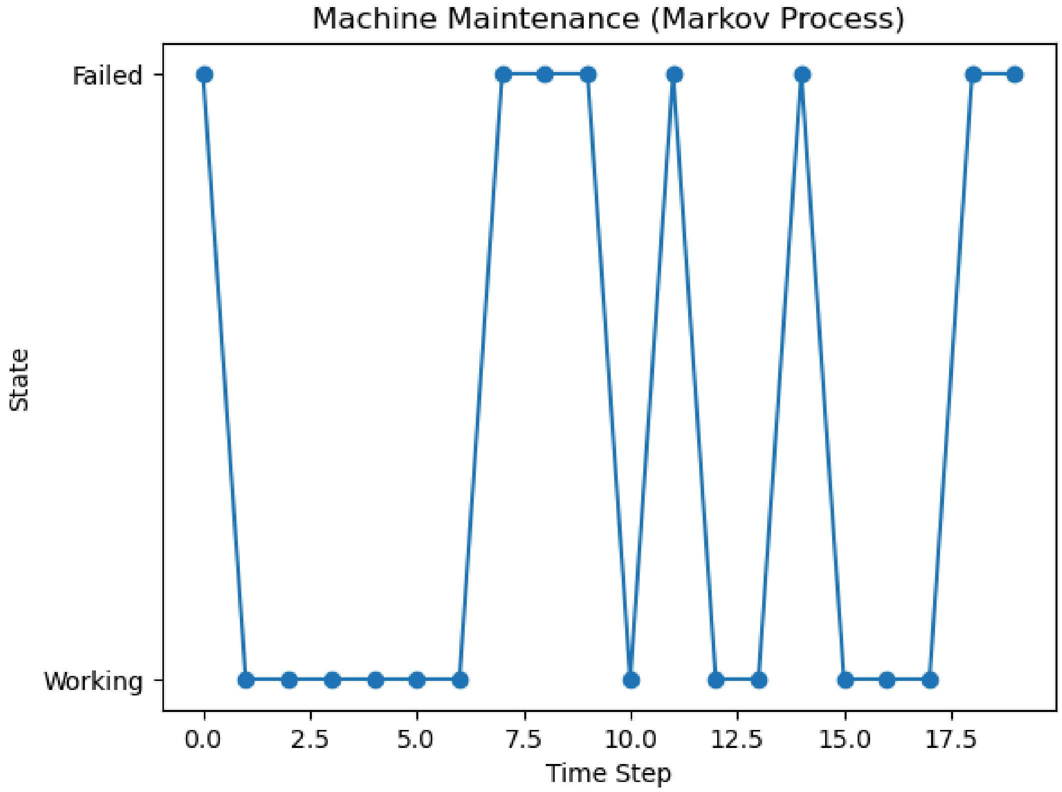

Machine Maintenance (Markov Process)
Failed
Working
State
0
2
4
6
8
10
12
14
16
Time Step

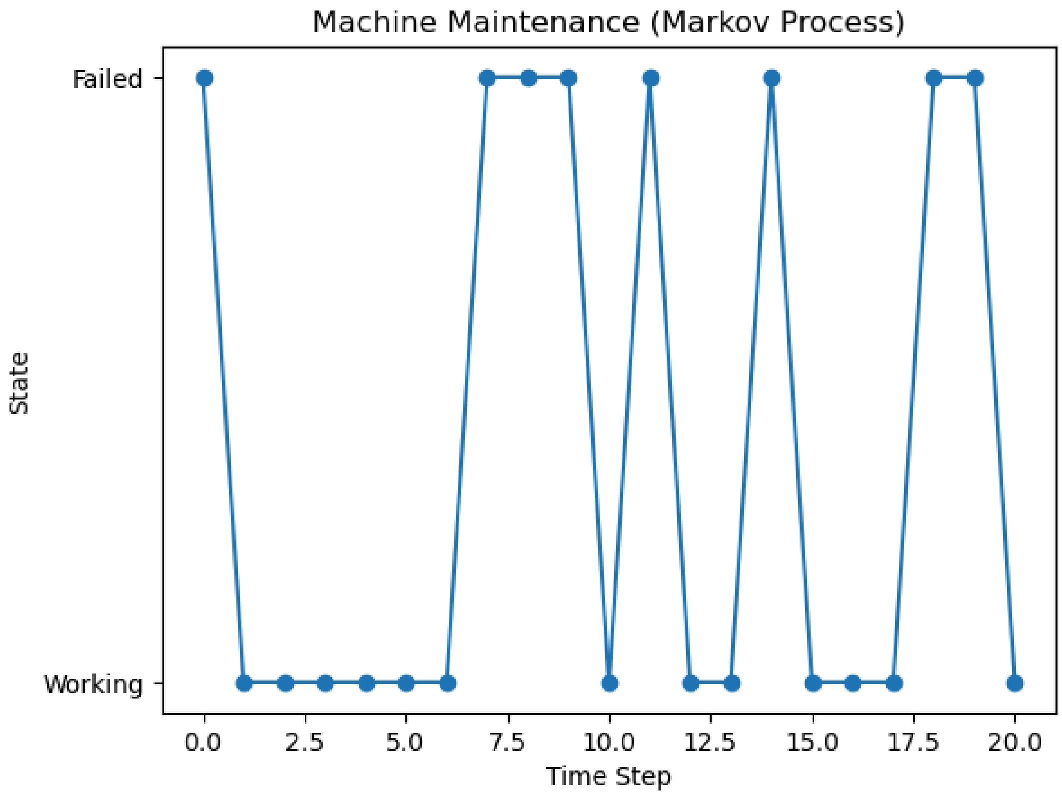


Machine Maintenance (Markov Process)
Failed
Working
State
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time Step



Machine Maintenance (Markov Process)
Failed
Working
State
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0.0
2.5
5.0
7.5
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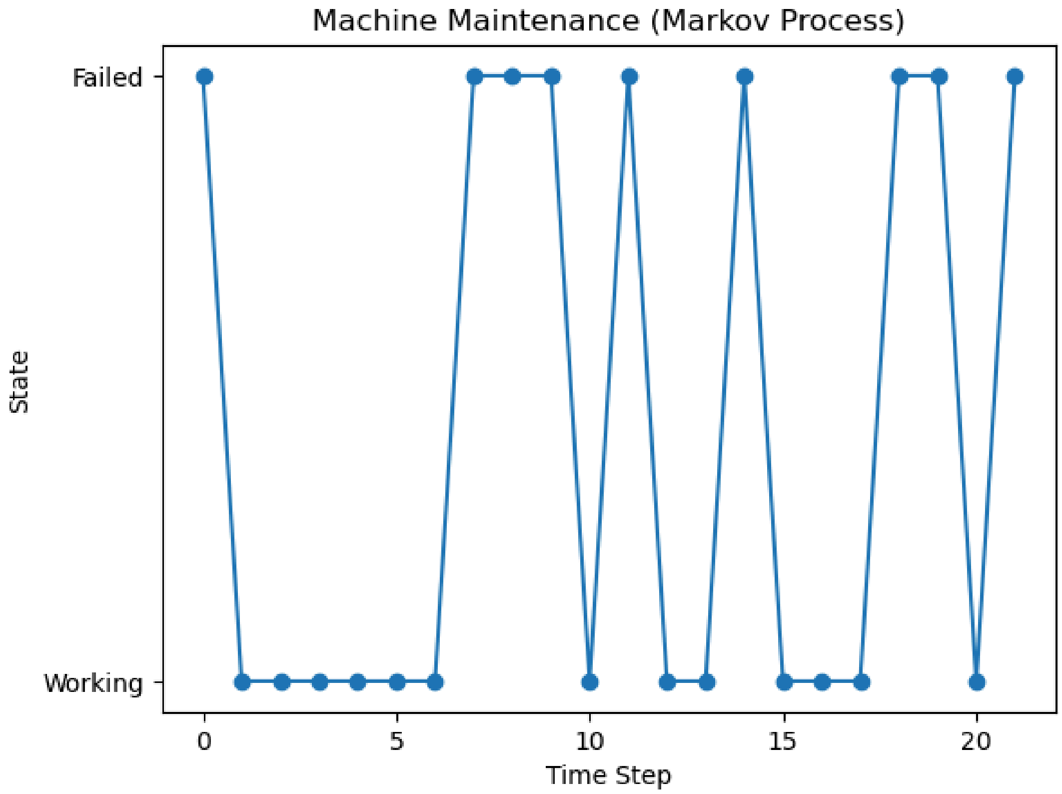
12.5
15.0
17.5
20.0
Time Step


Machine Maintenance (Markov Process)
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Working
State
0
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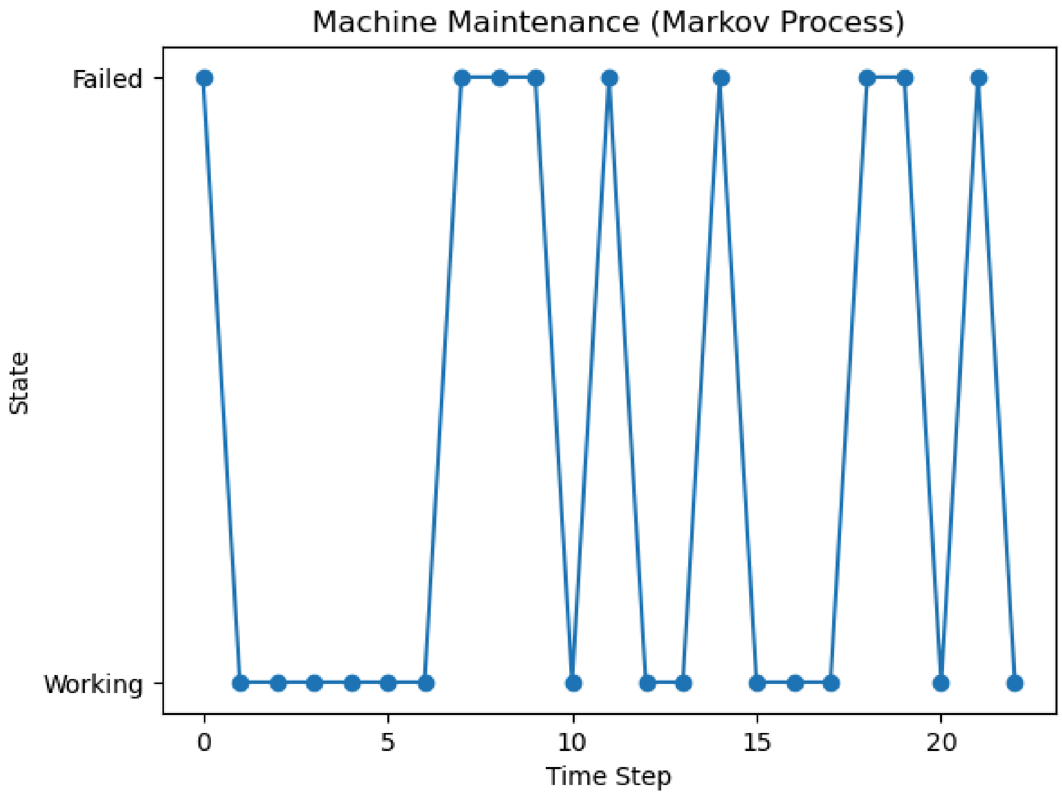
15
20
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
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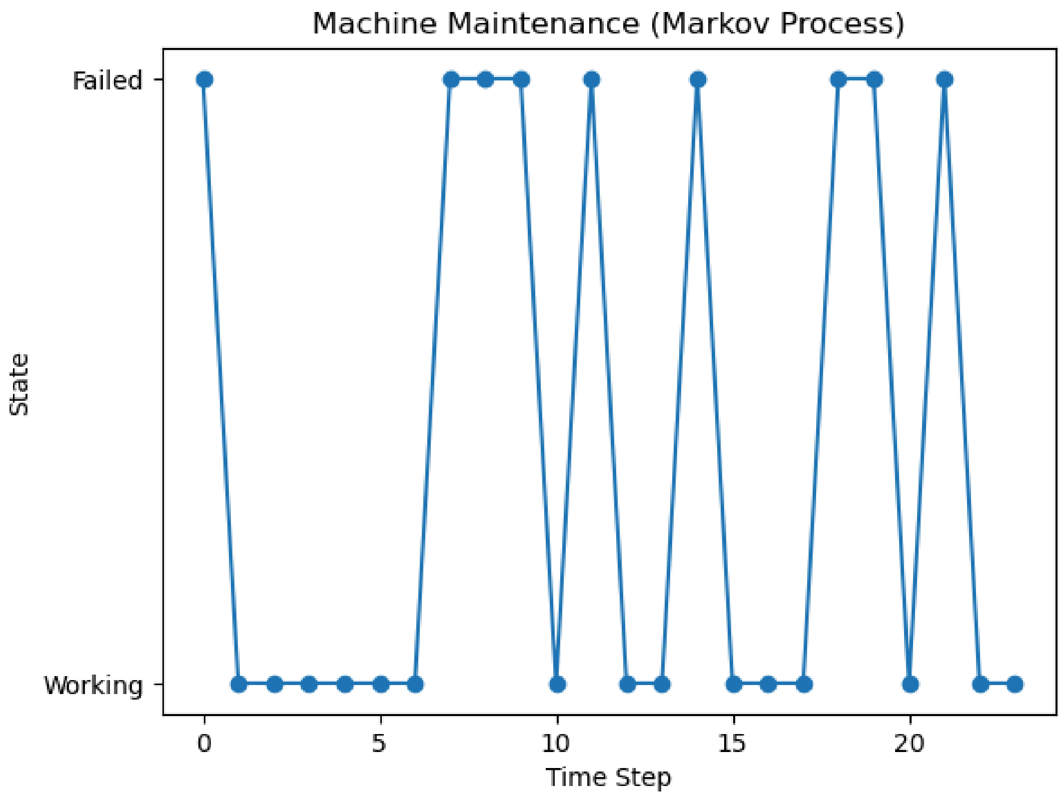
20
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
5
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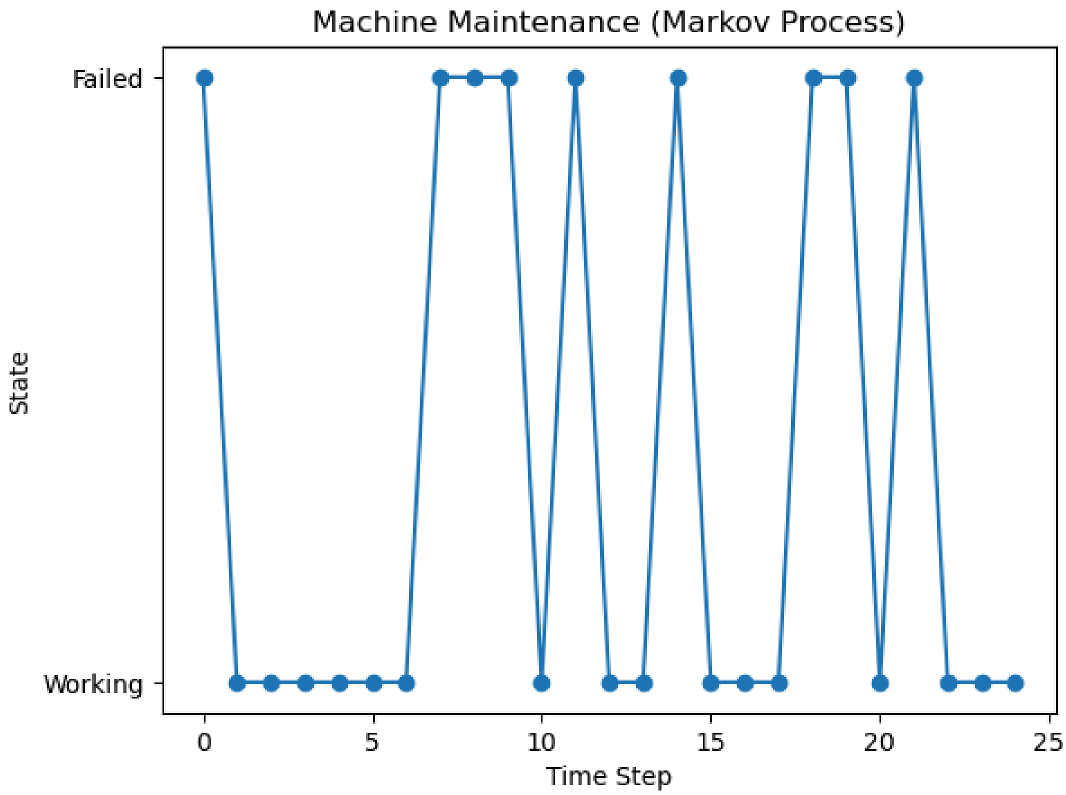
15
20
Time Step


Machine Maintenance (Markov Process)
Failed
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State
0
5
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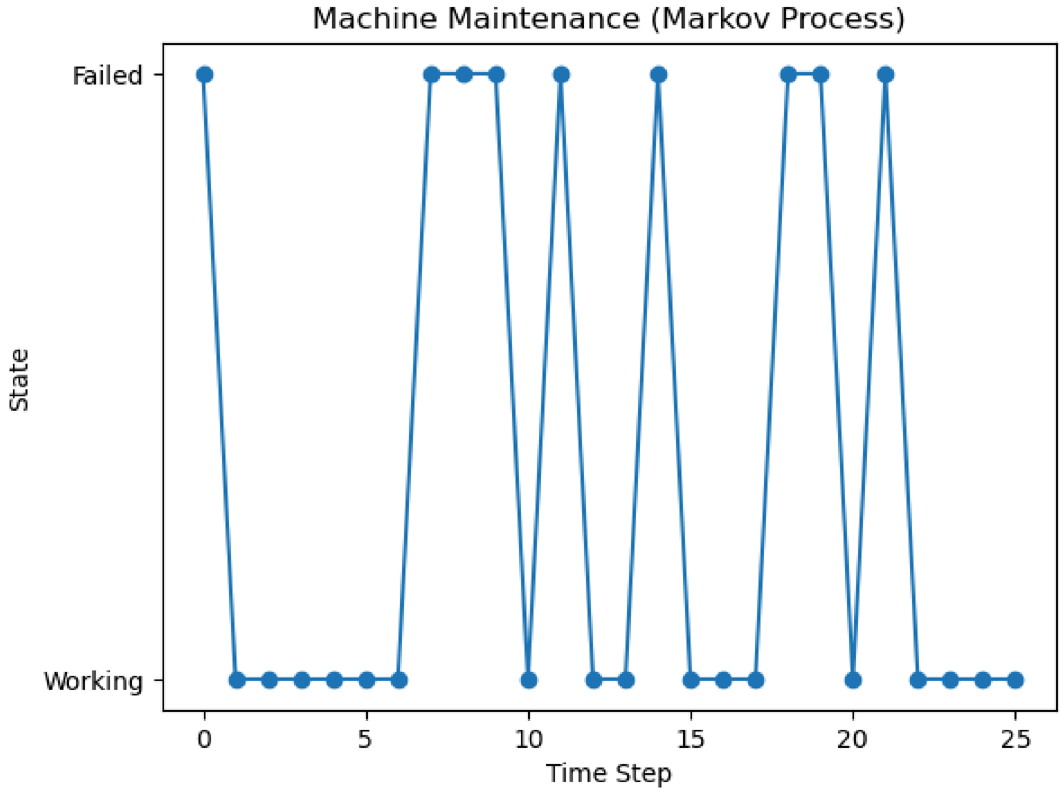
15
20
25
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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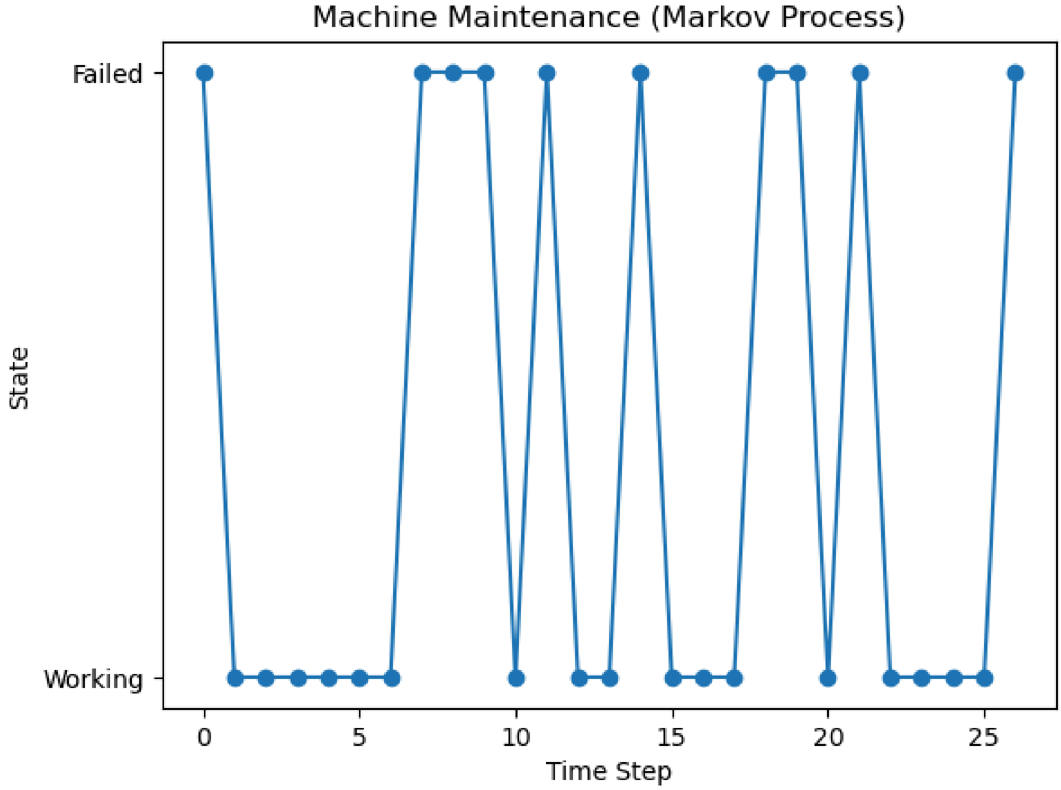
Time Step


Machine Maintenance (Markov Process)
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0
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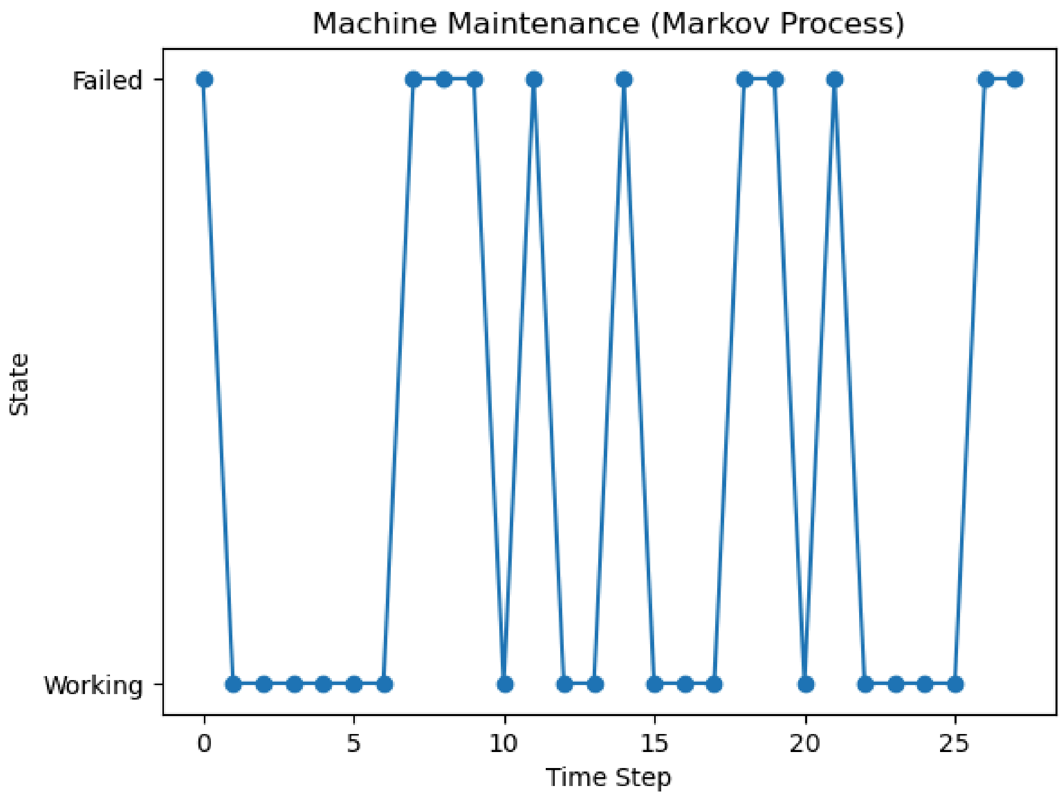
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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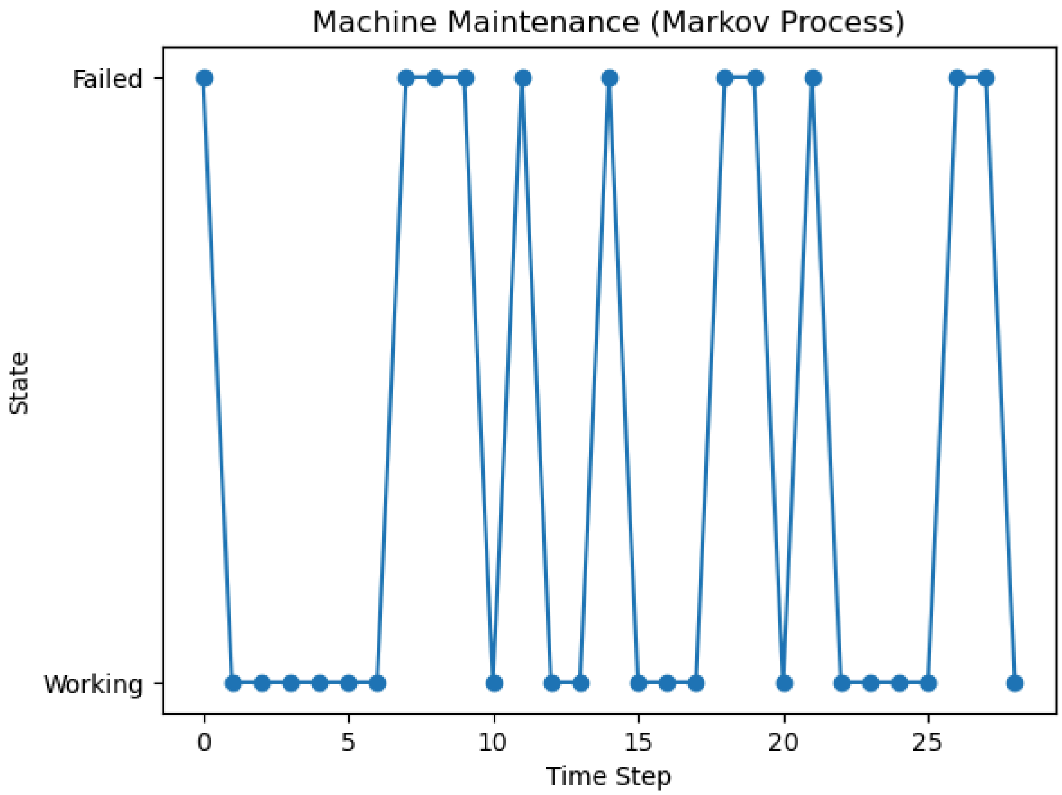
25
Time Step


Machine Maintenance (Markov Process)
Failed
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State
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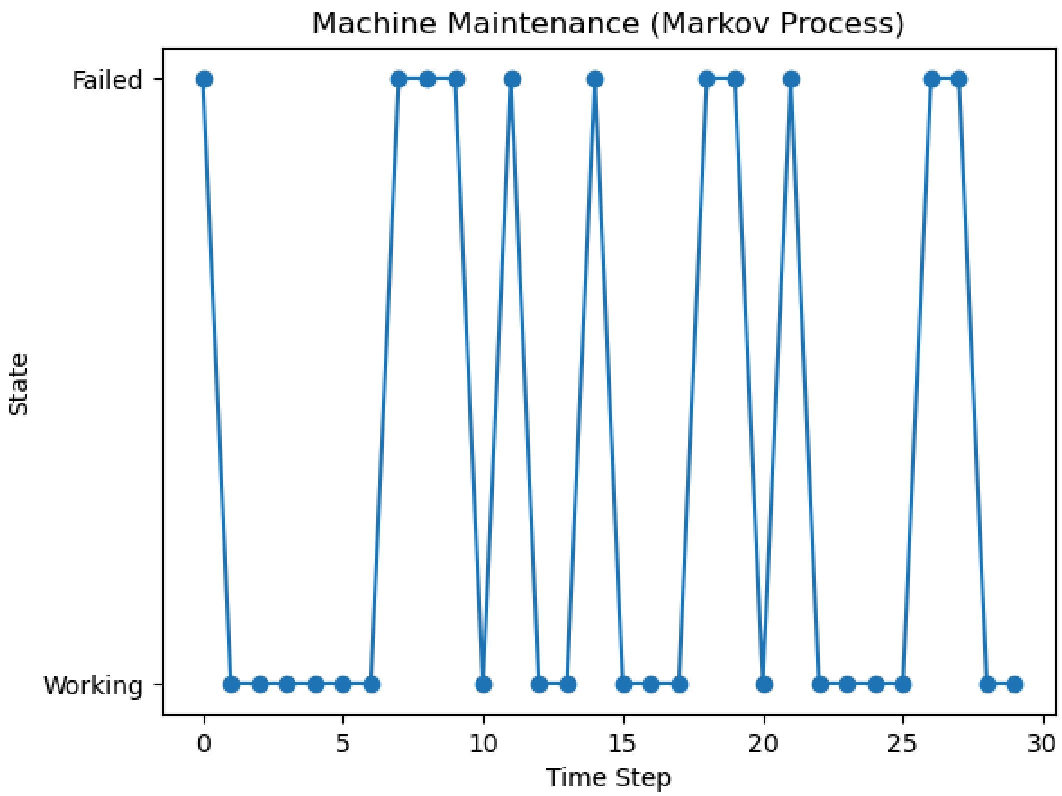
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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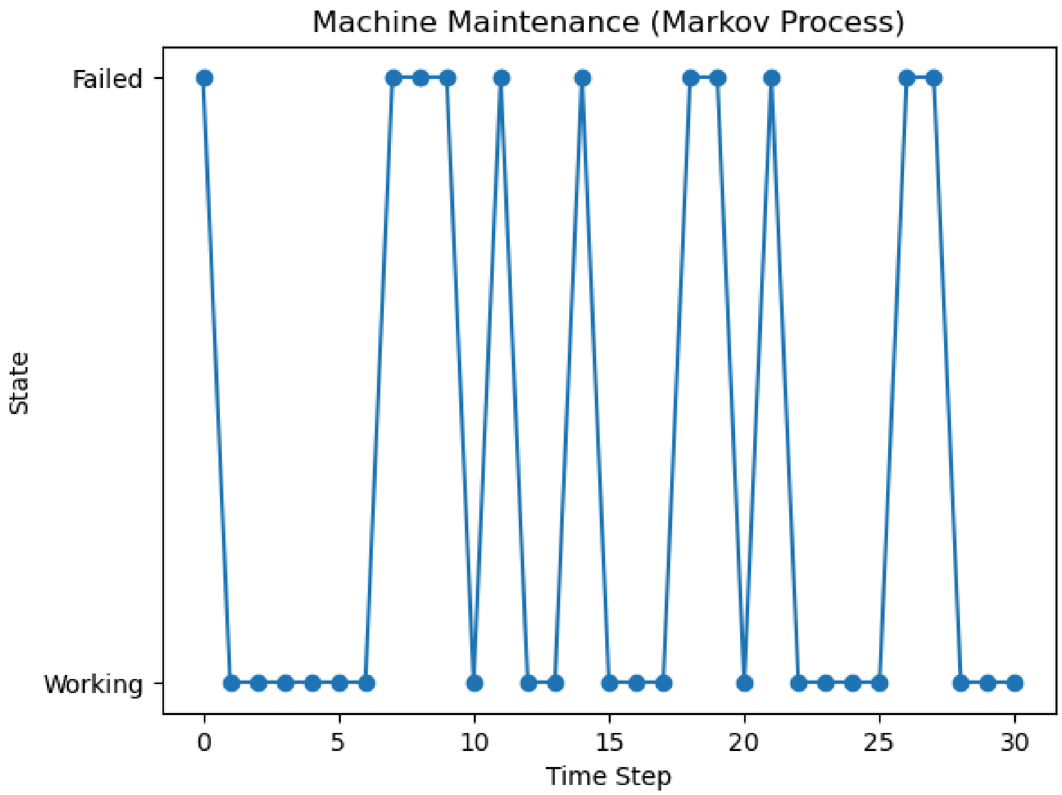
Time Step


Machine Maintenance (Markov Process)
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Working
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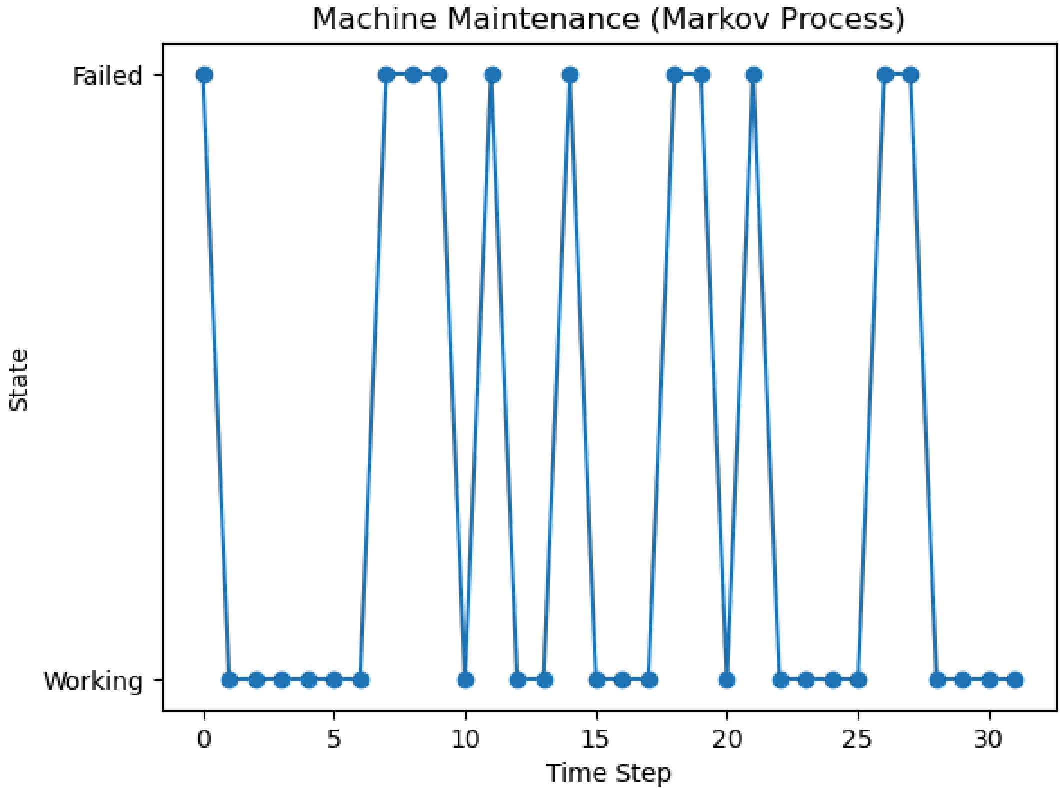
Time Step


Machine Maintenance (Markov Process)
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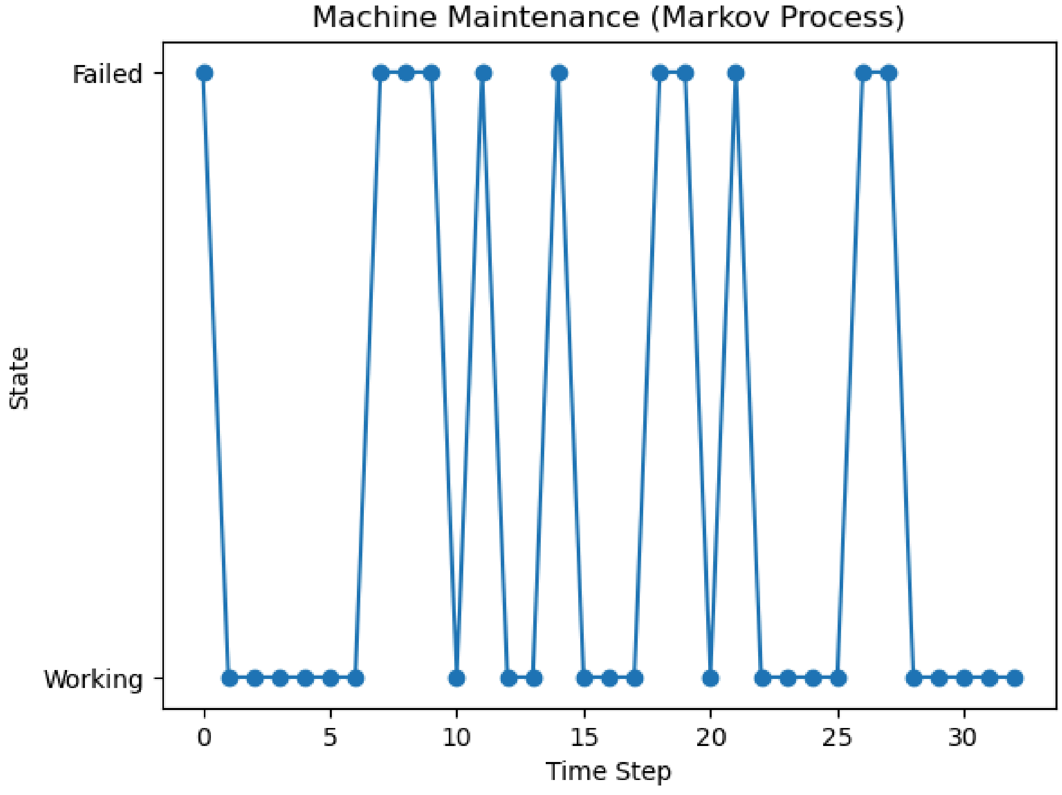
Time Step


Machine Maintenance (Markov Process)
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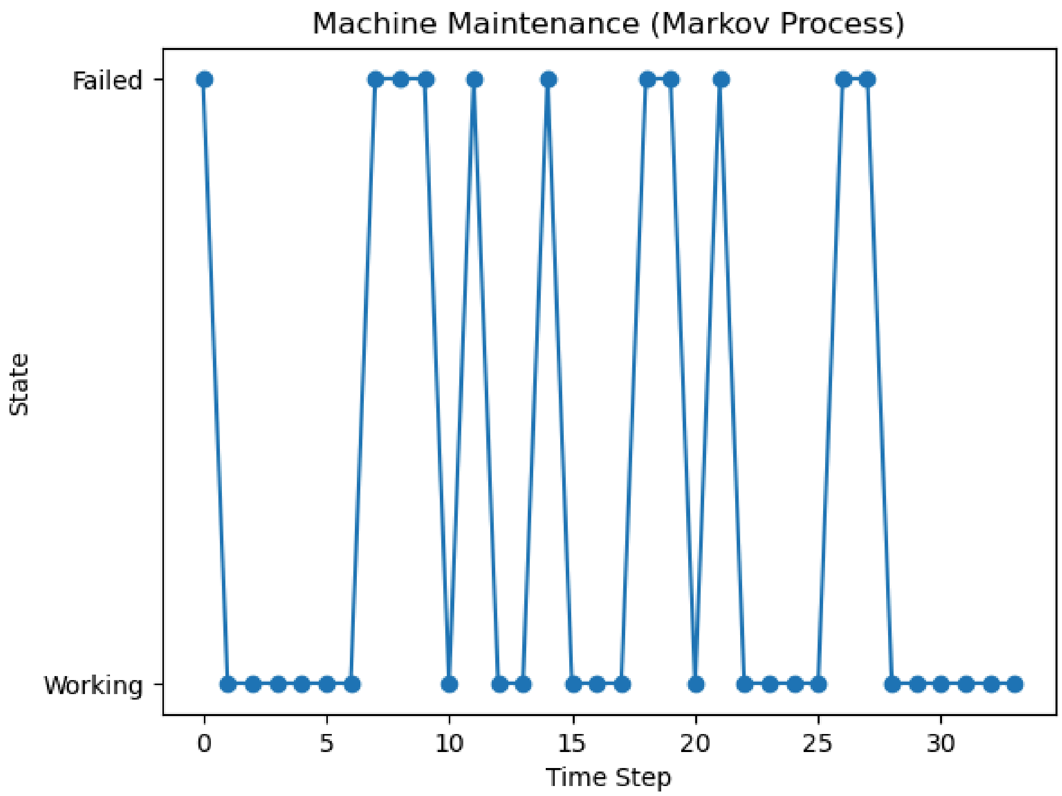
Time Step


Machine Maintenance (Markov Process)
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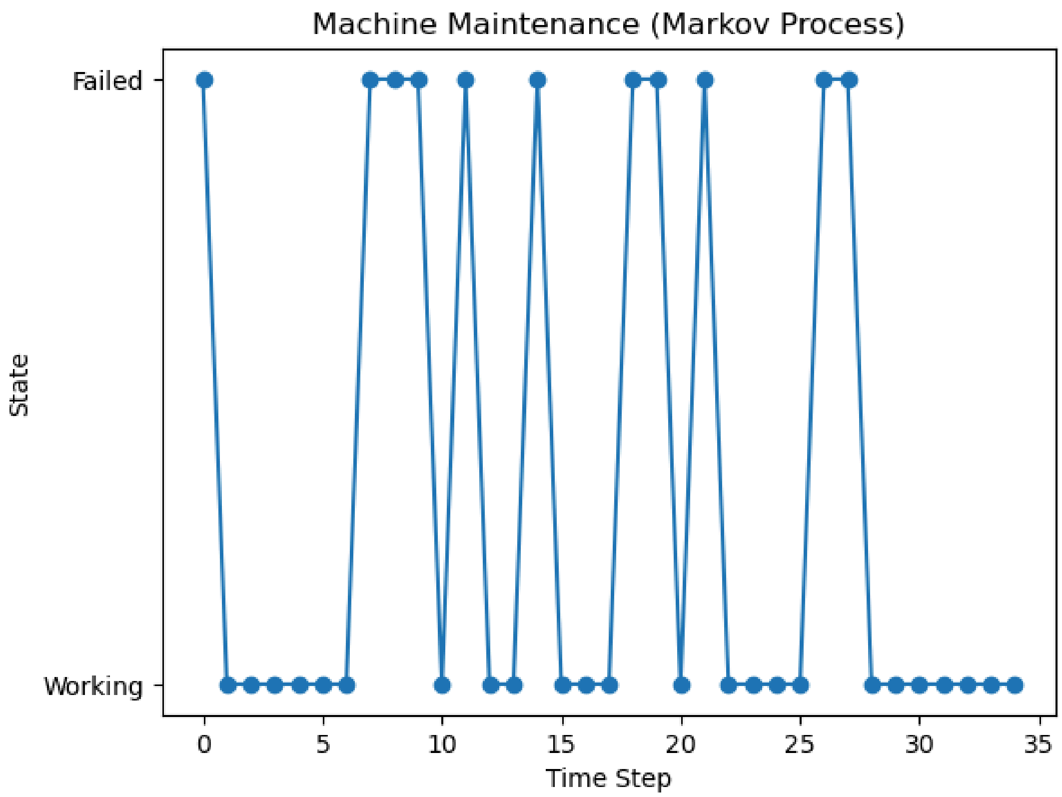
Time Step


Machine Maintenance (Markov Process)
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State
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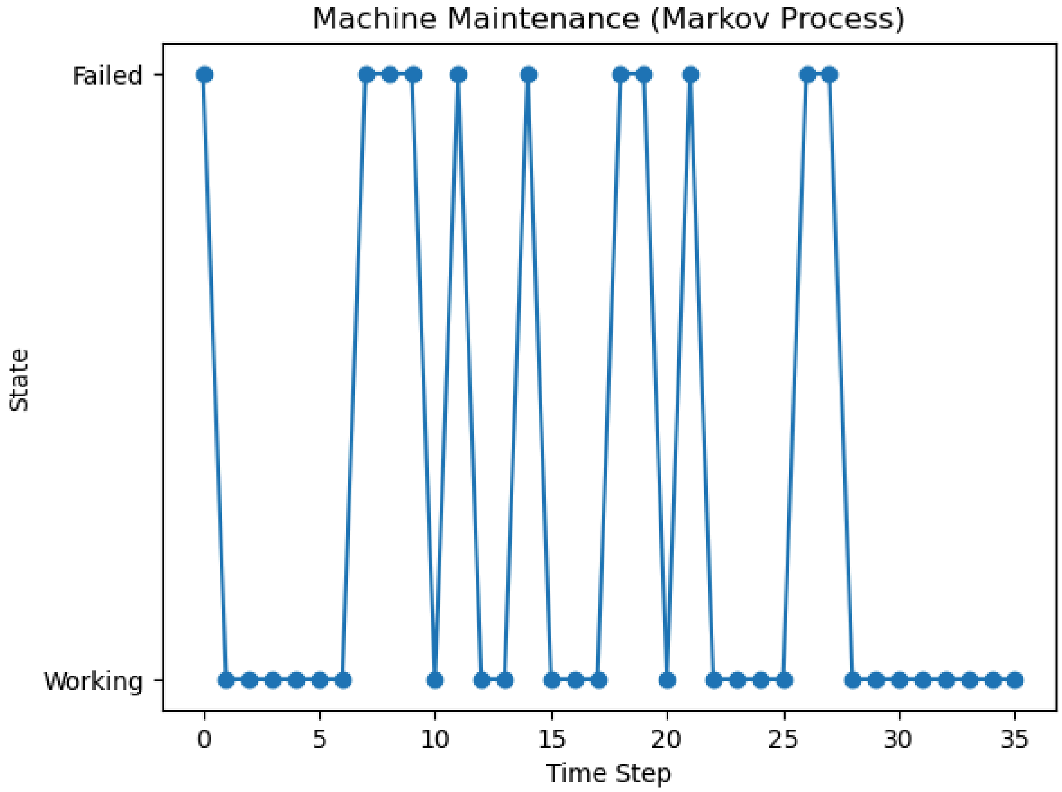
Time Step



Machine Maintenance (Markov Process)
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State
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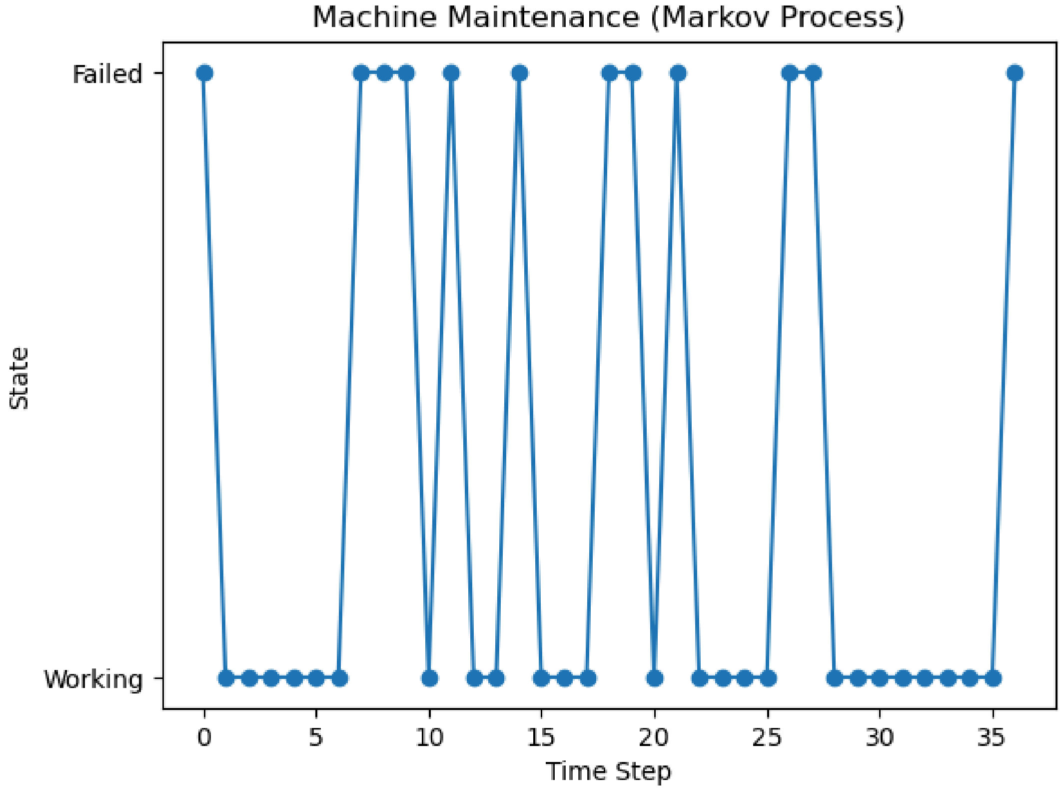
Time Step



Machine Maintenance (Markov Process)
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Working
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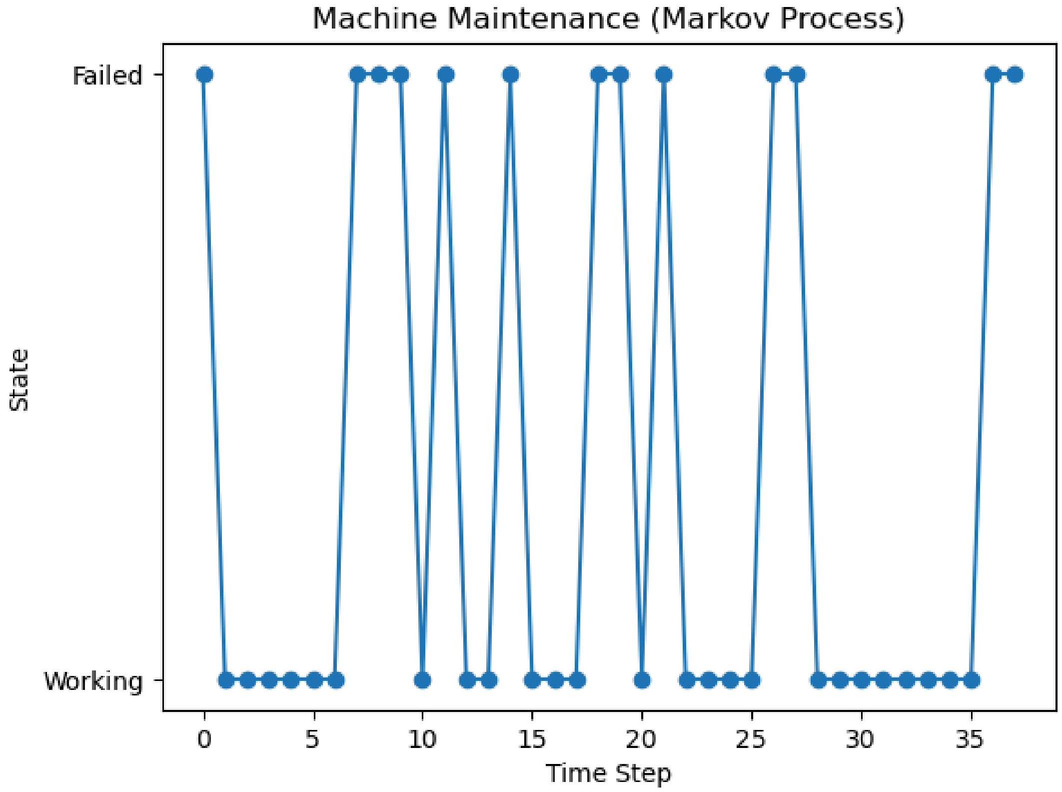
Time Step


Machine Maintenance (Markov Process)
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Working
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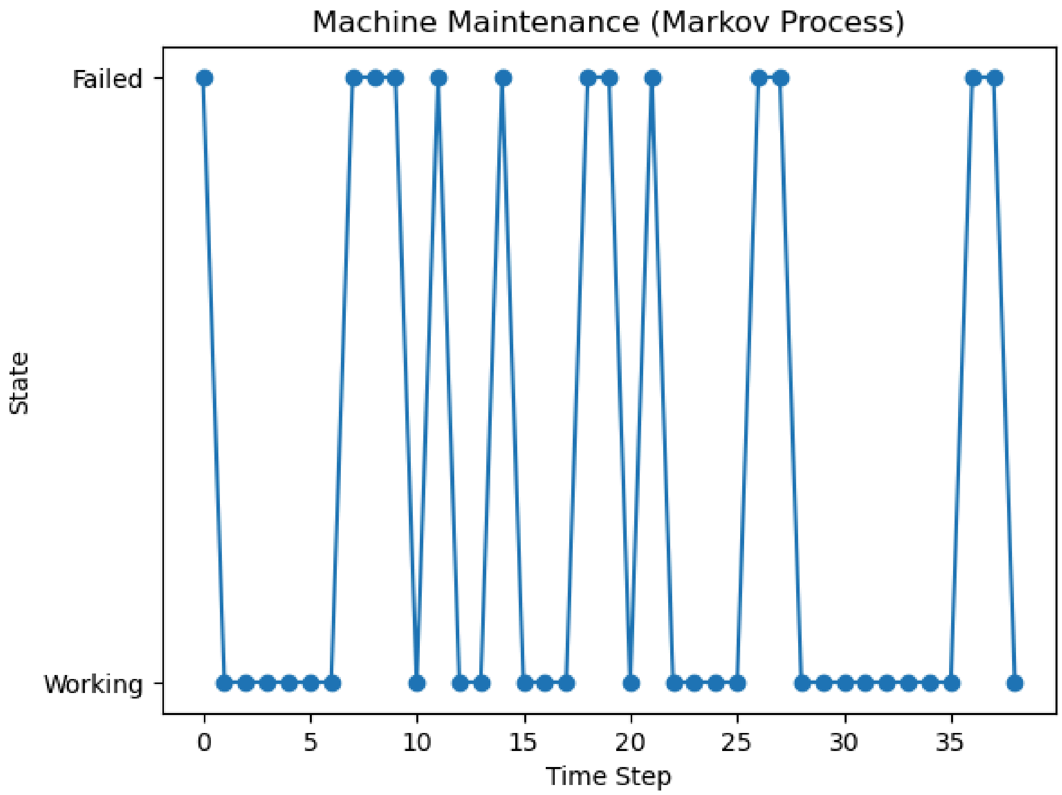
Time Step


Machine Maintenance (Markov Process)
Failed
Working
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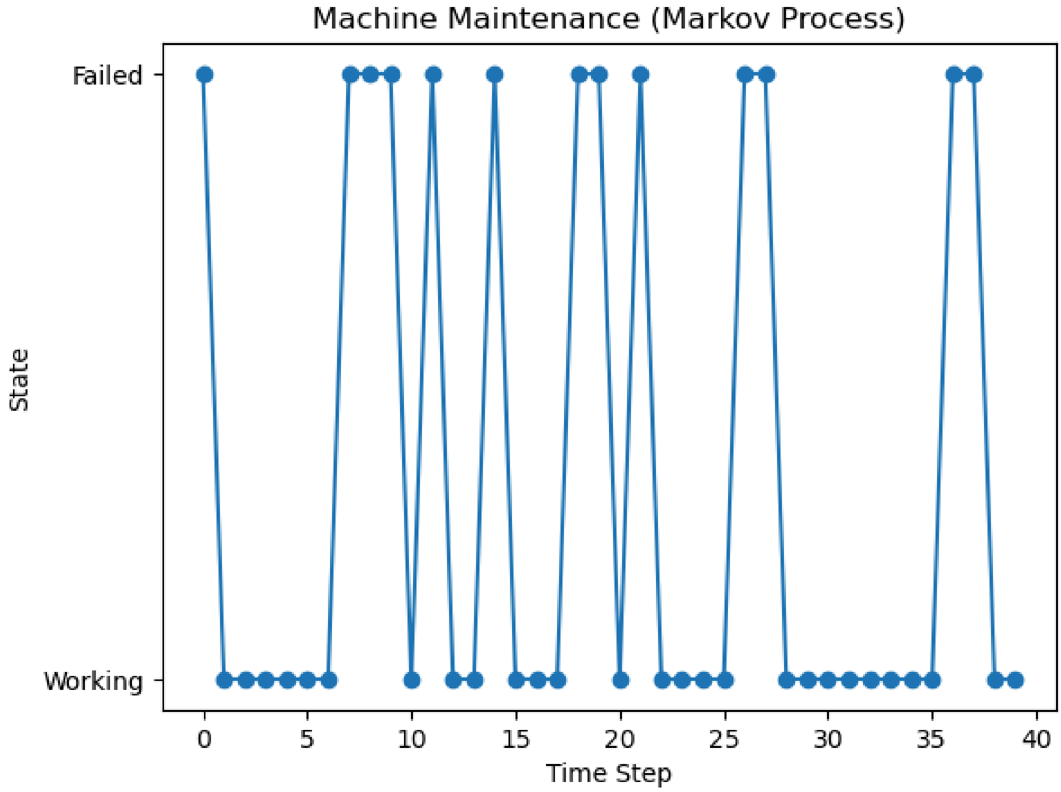
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
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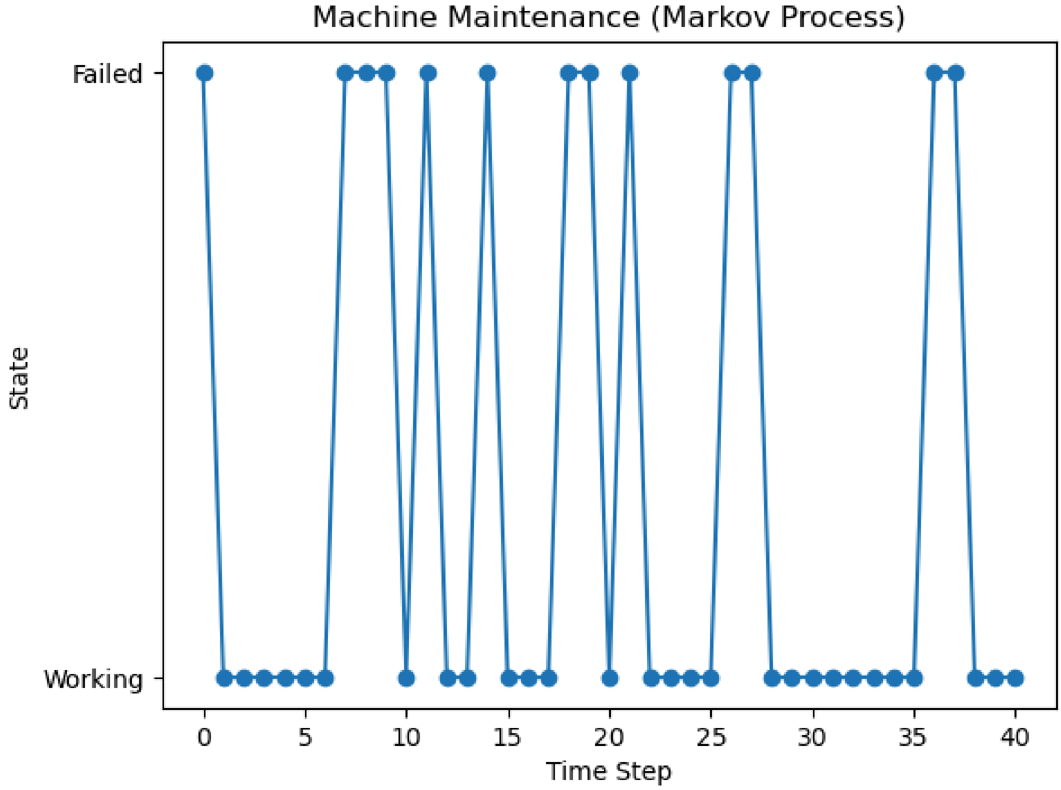
Time Step


Machine Maintenance (Markov Process)
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Working
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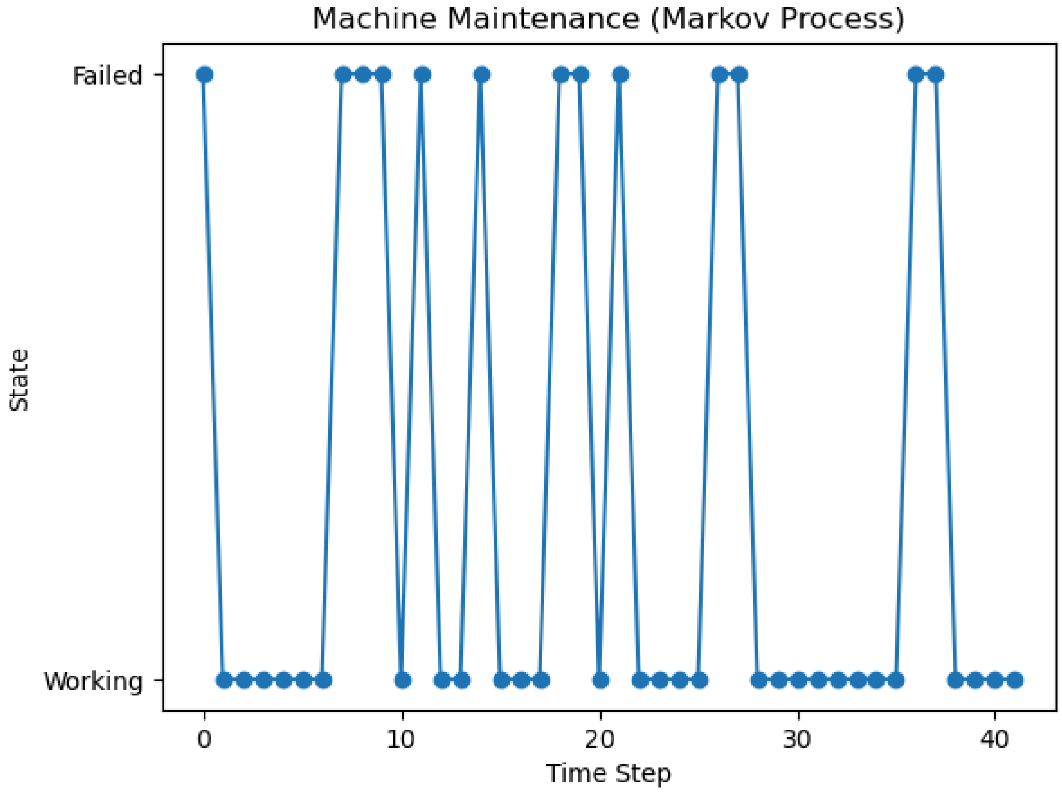
Time Step


Machine Maintenance (Markov Process)
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Working
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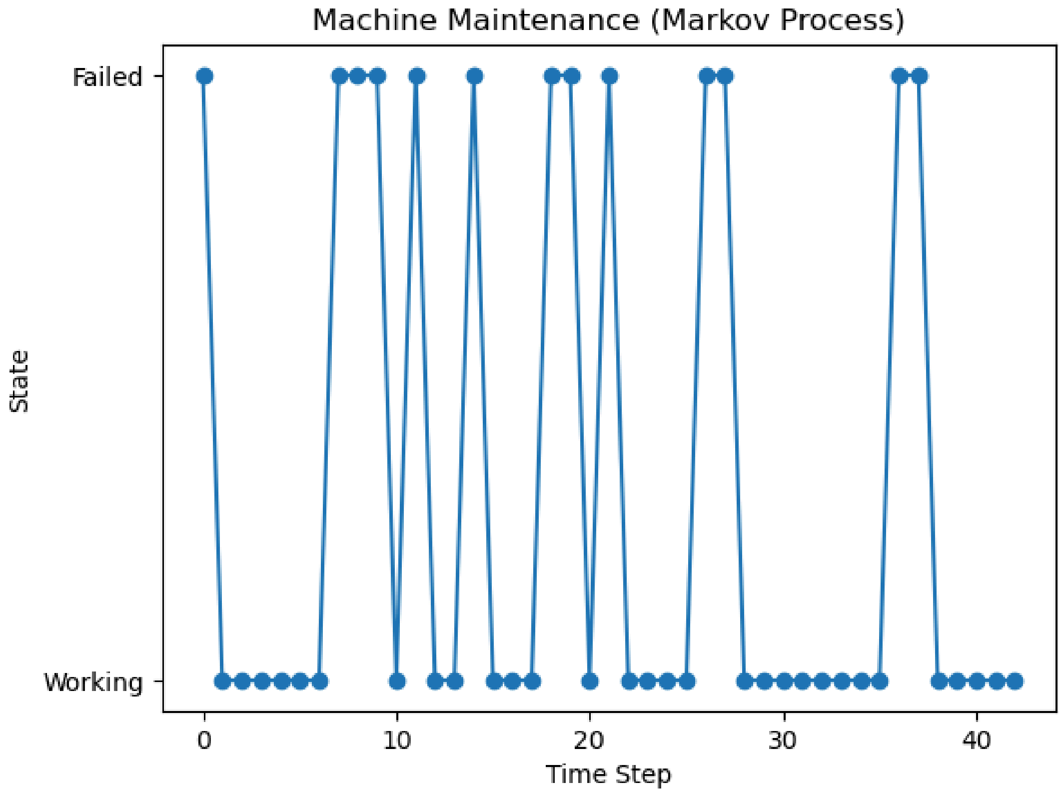
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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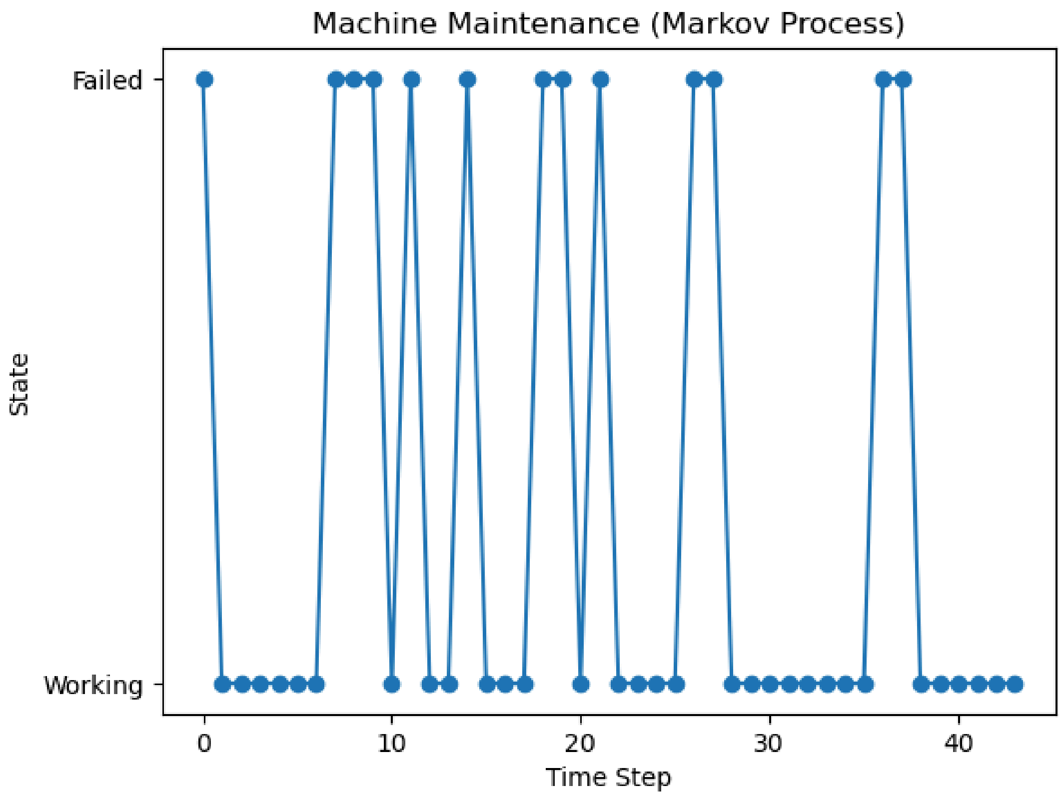
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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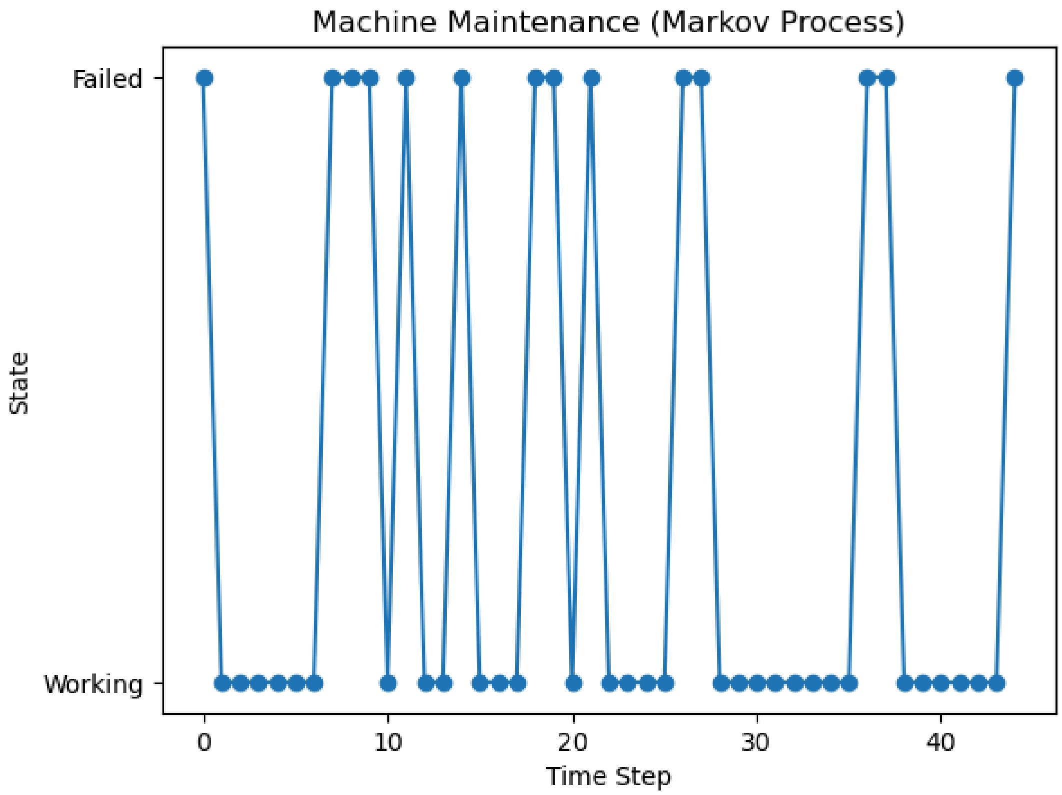
40
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
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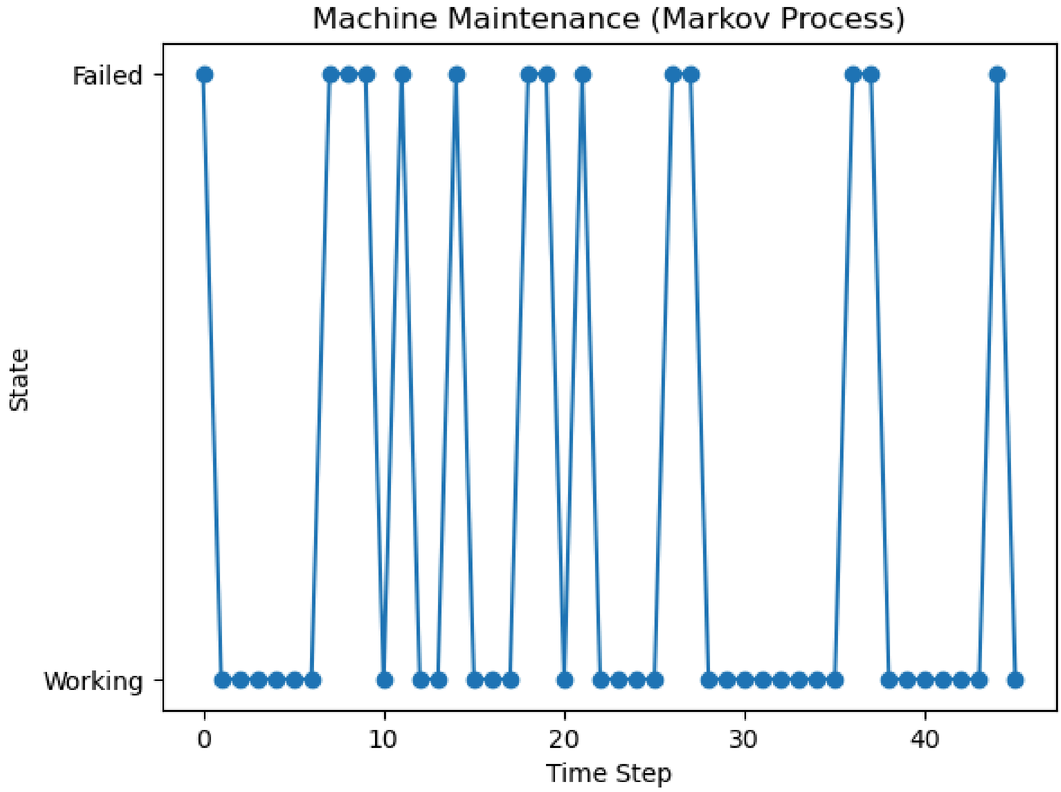
30
40
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
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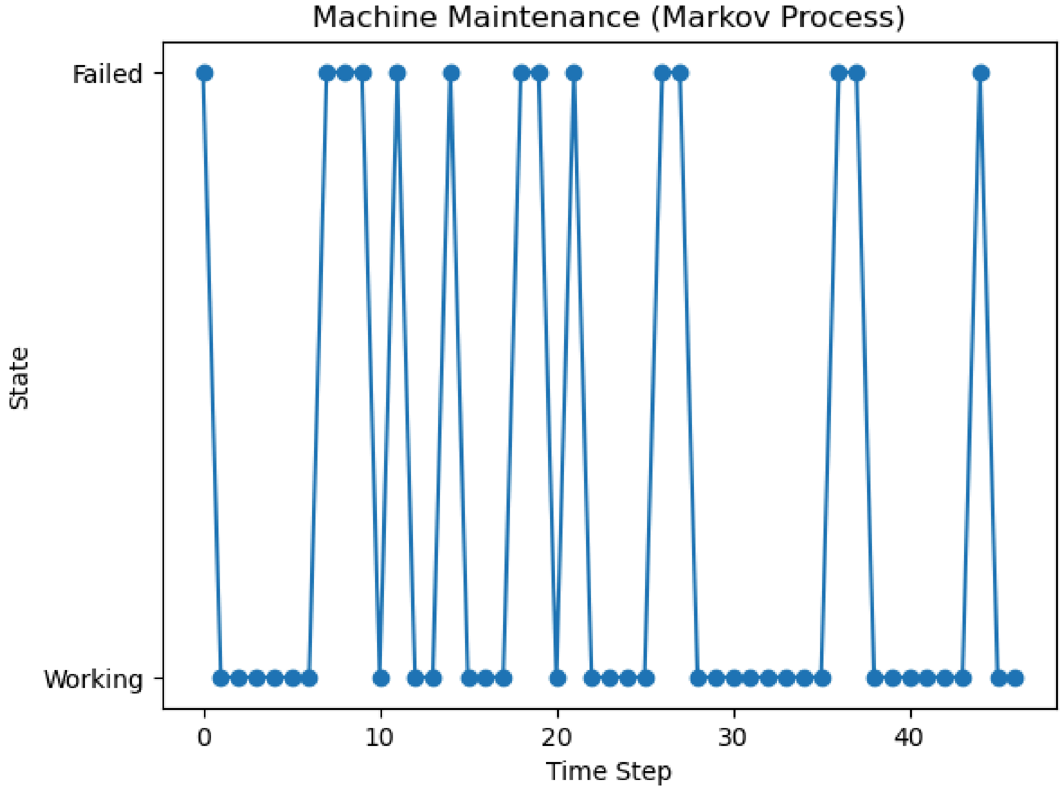
30
40
Time Step


Machine Maintenance (Markov Process)
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Working
State
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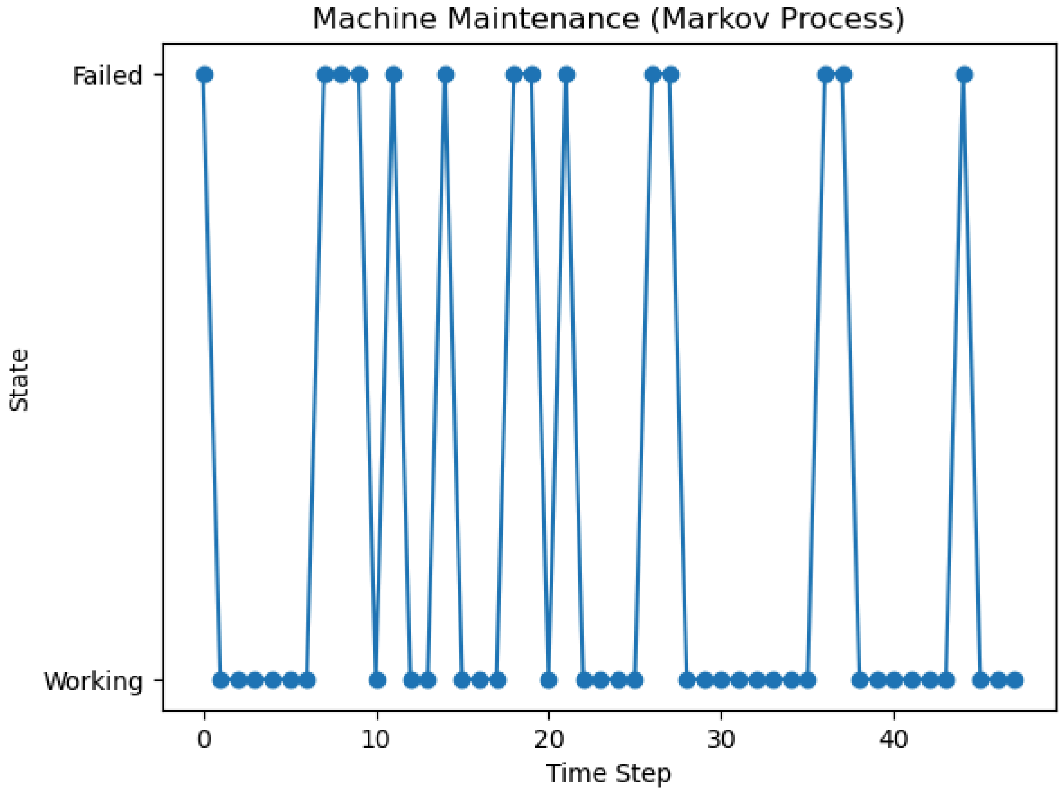
40
Time Step


Machine Maintenance (Markov Process)
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State
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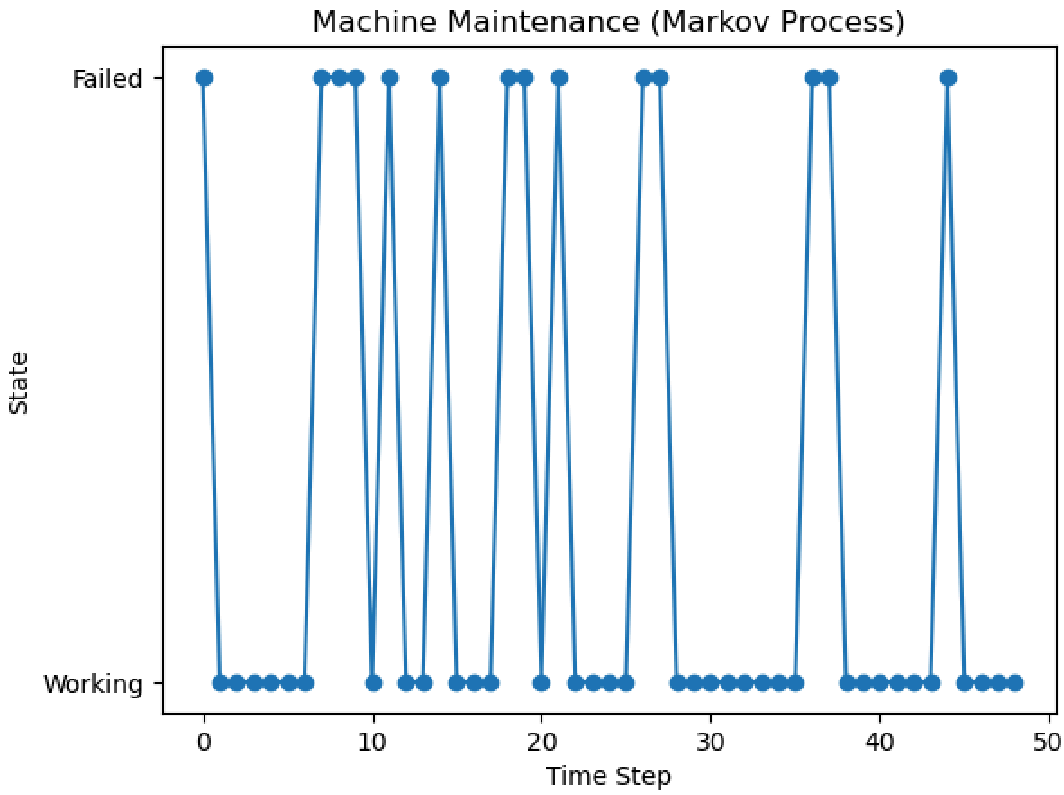
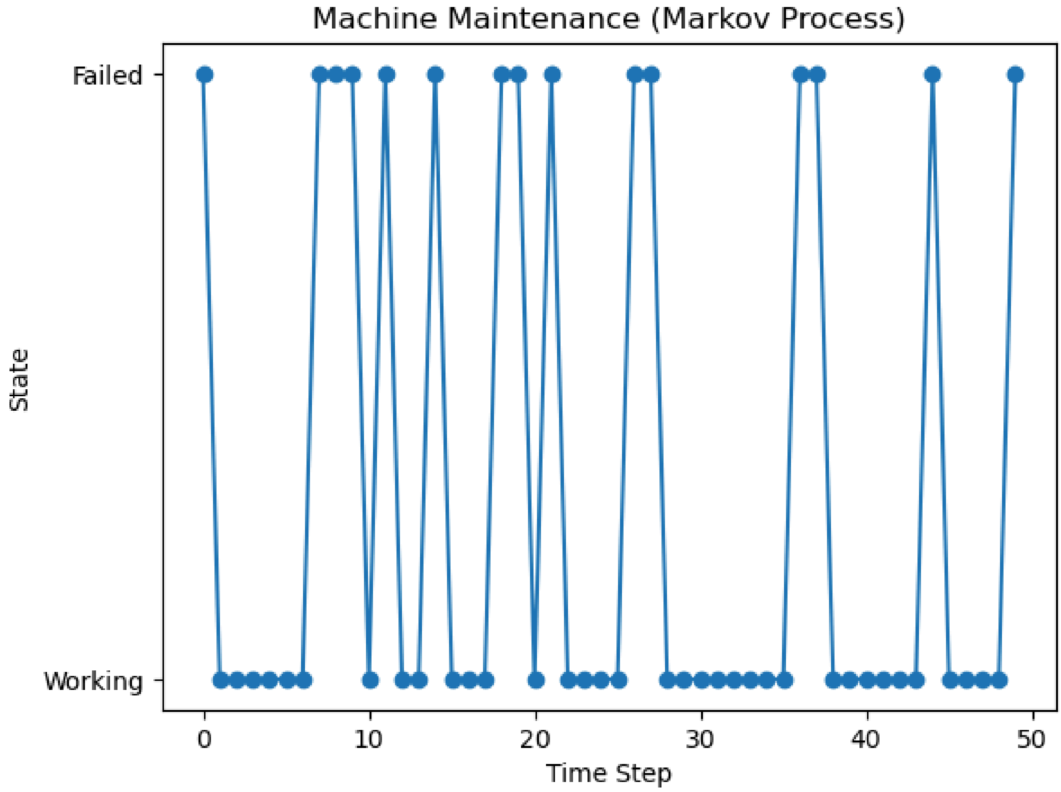
Time Step


Machine Maintenance (Markov Process)
Failed
Working
State
0
10
20
30
40
50
Time Step



In [ ]:
```
SE-A,S3
ROLL NO -341
```